COLLEGE TIME TRACKER (CTT)

Project Supervisor

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Submitted By

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BSCS 2023-25

CERTIFICATE

This is to certify that **Kainat Najamudin (2023-SD), BSCS 2023-25** has worked on and completed their Software Project at Department of Computer Science, **Government Graduate College, Sadiq Abad** in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science (BSCs) under my guidance and supervision.

In our opinion, it is satisfactory, up to the mark, and therefore fulfills the requirements of Bachelor of Science in Computer Science **(BSCS)**.

Supervisor / Internal Examiner

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External Examiner/Subject Specialist

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(Signature)

Accepted By:

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(For office use)

DEDICATION

"This project is devoted to my elder brother, whose constant encouragement and selfless support have been my greatest motivation. His belief in my abilities has always pushed me to move forward with confidence. I owe much of my progress to his guidance, strength, and the silent sacrifices he has made for my success."

ACKNOWLEDGEMENT

*I extend my heartfelt gratitude to my respected supervisor, **Mr. Sajjad Abdullah**, for his continuous guidance, encouragement, and valuable insights throughout the development of this project. His expertise and patience have been a constant source of motivation and learning. I sincerely appreciate his time, support, and constructive feedback, which played a vital role in the successful completion of my project, College Time Tracker.*

Abstract

The College Time Tracker (CTT) is a web-based academic management system designed to eliminate the manual and error-prone process of creating and distributing class timetables in academic institutions. Traditional scheduling methods rely on spreadsheets and manual coordination between departments, which frequently result in overlapping classes, double-booked rooms, and teacher scheduling conflicts. CTT addresses this problem by providing a centralized, role-based platform where an Administrator, Academic In-Charge, Teacher, and Student each interact through dedicated dashboards suited to their responsibilities.

The system allows authorized staff to manage user accounts, academic programs, classes, subjects, sections, rooms, and periods, and to build a weekly timetable through an interactive grid builder. A core feature of the system is its automated conflict-detection engine, which cross-checks every new timetable entry against existing entries to prevent a teacher from being scheduled in two classes during an overlapping time period.

The system is built using PHP for server-side logic, MySQL/MariaDB for relational data storage, and HTML5, CSS3, Bootstrap 5, and JavaScript for a responsive front-end. Passwords are stored using secure one-way hashing, and access to each module is restricted using role-based access control. The final system was tested using black-box testing techniques covering positive, negative, and boundary test cases, achieving a high pass rate across all functional modules.

The outcome is a lightweight, reliable, and easy-to-use academic scheduling platform suitable for colleges and small universities, reducing administrative workload and eliminating scheduling conflicts.

Keywords: Timetable Management, Web Application, MySQL, Role-Based Access Control, Conflict Detection

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CHAPTER 1:

INTRODUCTION

1.1 Background of the Study

Domain Overview: Academic scheduling is one of the most repetitive yet critical administrative functions in any college. It involves assigning subjects, teachers, rooms, and time periods to multiple classes across different programs and semesters, while avoiding overlaps.

Current System: In most colleges, timetables are currently prepared manually using spreadsheet software such as Microsoft Excel. The Academic In-Charge collects teacher availability, room availability, and subject load individually and constructs the timetable manually before printing/distributing hard copies or PDF files.

Existing Problems:

- Manual timetables frequently contain human errors, resulting in double-booked teachers or rooms.
- There is no centralized system for teachers or students to view their personal schedule online.
- Updates to the timetable are not reflected in real time, causing confusion.
- Historical records of previous timetables are not properly maintained.
- There is no automated way to generate reports such as teacher workload.

1.2 Problem Statement

There is no centralized, automated system in the college that allows academic staff to create, manage, and distribute class timetables while automatically preventing scheduling conflicts, and that allows teachers and students to view their personal schedules through a secure, role-based online platform.

1.3 Objectives

General Objective:

To design and develop a web-based College Time Tracker system that automates the creation, management, and distribution of academic timetables while preventing scheduling conflicts.

Specific Objectives:

- Develop a secure, role-based login system for four types of users.
- Design a relational database to store users, academic entities, and timetable data.

- Implement a module to manage academic entities (Programs, Classes, Subjects, Sections, Rooms, Periods).
- Develop an interactive timetable builder with a grid-based interface.
- Implement an automated conflict-detection algorithm to prevent double-booking of teachers.
- Provide teachers and students with a personalized dashboard to view their schedule.
- Generate workload and scheduling reports for administrative use.
- Evaluate the system through structured black-box testing.

1.4 Scope of the Project

Included Features:

- User management for Administrator, Academic In-Charge, Teacher, and Student.
- Management of Programs, Classes, Subjects, Sections, Rooms, and Periods.
- Interactive timetable builder with automated conflict prevention.
- Role-based dashboards and access control.
- Teacher workload and class-load reports.
- CSV export and print-friendly timetable view.

Excluded Features:

- Mobile native application (Android/iOS).
- SMS/Email notification integration.
- Payment or fee management modules.
- Multi-campus/multi-branch synchronization.

1.5 Significance of the Project

Academic Value: The project provides hands-on experience in database design, authentication, role-based access control, and algorithm design (conflict detection), reinforcing concepts learned in Software Engineering and Database Systems courses.

Industry Value: Small and medium colleges that cannot afford commercial ERP systems can adopt a lightweight system such as CTT to digitize their scheduling process at low cost.

Beneficiaries: Administrators, Academic In-Charge staff, Teachers, and Students all directly benefit from reduced manual effort and improved schedule accuracy.

1.6 Overview of Proposed System

CTT is a centralized web application in which an Administrator creates Academic In-Charge accounts; the Academic In-Charge in turn manages Teacher and Student accounts and constructs the timetable using a grid builder that maps Classes (rows) against Periods (columns). Each cell may contain one or more entries (Subject, Teacher, Room, Section, Days). Before an entry is saved, the system automatically checks whether the selected teacher is already assigned to another class during an overlapping day/period combination; if a conflict is found, the entry is rejected with a clear error message. Teachers and Students log in to view only their own relevant schedule.

1.7 Tools and Technologies

Category	**Technology**
Programming Language	PHP 8.x, JavaScript (ES6)
Frontend	HTML5, CSS3, Bootstrap 5, Font Awesome
Database	MySQL / MariaDB
Server / Platform	Apache (XAMPP), Web Browser
IDE	Visual Studio Code
Diagramming	draw.io / Lucidchart

CHAPTER 2:

REQUIREMENTS SPECIFICATION

2.1 Overall Description

Product Perspective: CTT is a standalone, self-contained web application that does not depend on any external institutional system. It operates on a standard LAMP/WAMP stack.

Product Functions: User management, academic entity management, timetable construction, conflict prevention, and reporting.

User Characteristics: Administrators possess technical/system-management knowledge; Academic In-Charge staff have moderate computer literacy; Teachers and Students require only basic computer literacy to view schedules.

2.2 Functional Requirements

FR ID	Requirement	Actor
FR-01	The system shall allow the Administrator to log in and log out securely.	Administrator
FR-02	The system shall allow the Administrator to manage (create, update, delete) Academic In-Charge accounts.	Administrator
FR-03	The system shall allow the Administrator to manage the master timetable.	Administrator
FR-04	The system shall allow the Administrator to assign instructors to classes/courses.	Administrator
FR-05	The system shall allow the Administrator to generate and view system reports.	Administrator
FR-06	The system shall allow the Administrator to back up and restore the database.	Administrator
FR-07	The system shall allow the Academic In-Charge to log in and log out securely.	Academic In-Charge
FR-08	The system shall allow the Academic In-Charge to manage Teacher and Student accounts.	Academic In-Charge
FR-09	The system shall allow the Academic In-Charge to manage the timetable.	Academic In-Charge
FR-10	The system shall allow the Academic In-Charge to assign instructors to classes/courses.	Academic In-Charge

FR-11	The system shall allow the Academic In-Charge to generate and view reports.	Academic In-Charge
FR-12	The system shall allow the Teacher to log in and log out securely.	Teacher
FR-13	The system shall allow the Teacher to view their personal timetable.	Teacher
FR-14	The system shall allow the Student to log in and log out securely.	Student
FR-15	The system shall allow the Student to view their personal timetable.	Student
FR-16	The system shall automatically detect and prevent overlapping teacher assignments.	System
FR-17	The system shall allow authorized users to export the timetable to CSV and print it.	Admin / In-Charge

Table 2.1: Functional Requirements

2.3 Non-Functional Requirements

2.3.1 Performance Requirements

- The system shall load the timetable grid within 3 seconds under normal load.
- The conflict-check algorithm shall complete within 1 second per entry.

2.3.2 Security Requirements

- Passwords shall be stored using one-way hashing (bcrypt).
- Sessions shall expire after a period of inactivity.
- Each module shall enforce role-based access control.

2.3.3 Usability Requirements

- The interface shall be responsive across desktop, tablet, and mobile screens.
- Error messages shall be clear and actionable.

2.3.4 Reliability Requirements

- The system shall correctly persist all CRUD operations to the database without data loss.

2.3.5 Availability Requirements

- The system shall be available during standard operating hours (8 AM – 6 PM) with minimal downtime.

2.3.6 Scalability Requirements

- The database schema shall support the addition of new programs, classes, and users without structural change.

2.4 System Users

Role	Description
Administrator	Full system control; manages Academic In-Charge accounts and system settings.
Academic In-Charge	Manages Teacher/Student accounts and constructs the timetable.
Teacher	Views personal weekly teaching schedule.
Student	Views personal class timetable.

2.5 System Constraints

- Requires a web server supporting PHP 8.x and MySQL/MariaDB.
- Requires a modern web browser with JavaScript enabled.
- Limited to single-institution deployment (no multi-tenant support).

2.6 Use Case Diagram

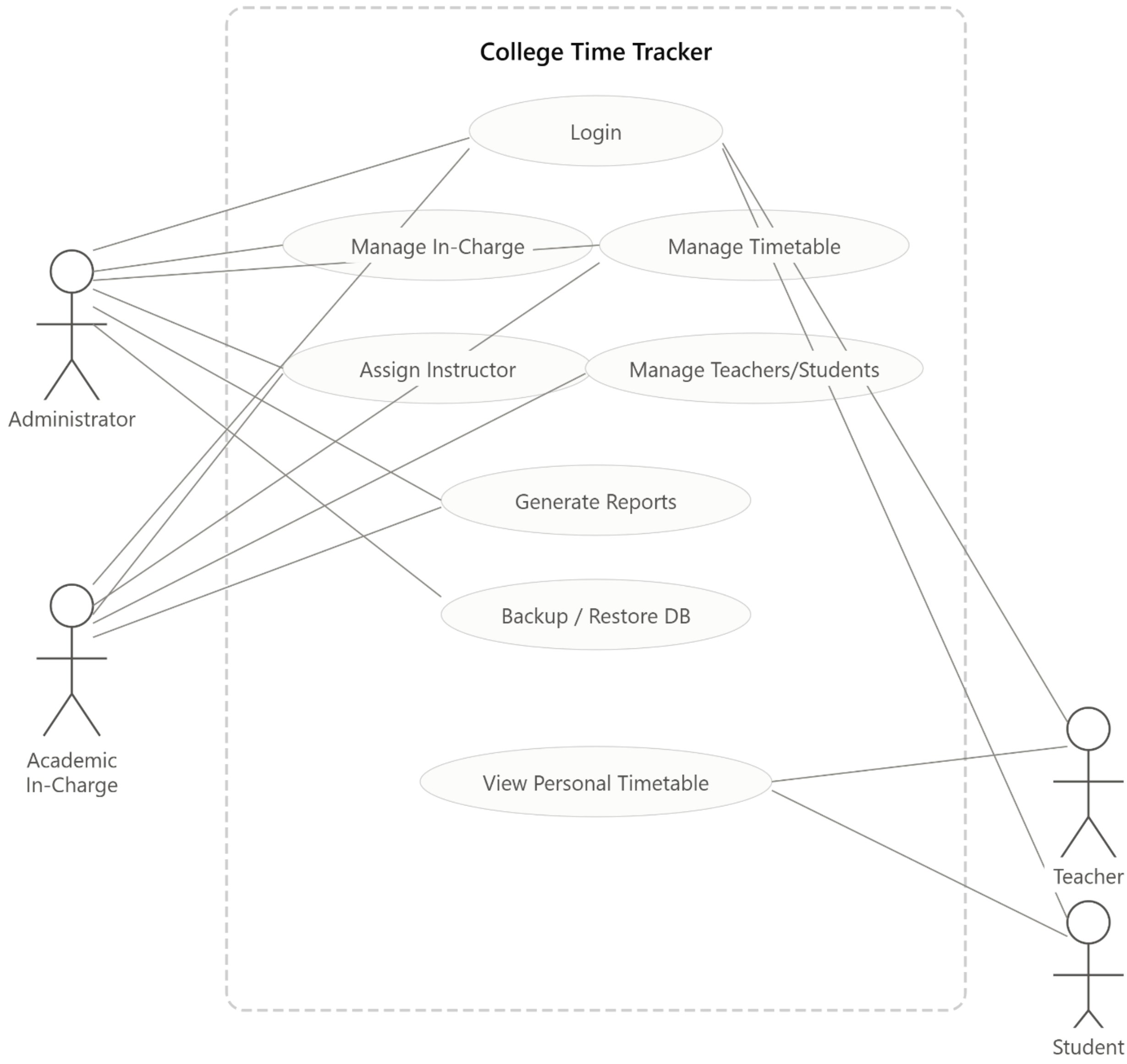


Figure 2.1, the system supports four actors, each with a distinct set of use cases.

2.7 Use Case Descriptions

Use Case: Login

Field	Detail
Use Case Name	Login
Actor	Administrator / Academic In-Charge / Teacher / Student

Field	Detail
Preconditions	User must already have a registered account.
Main Flow	1. User opens login page. 2. User enters username and password. 3. System validates credentials against the database. 4. System redirects the user to the role-specific dashboard.
Alternate Flow	If credentials are invalid, the system displays an error message and remains on the login page.
Postcondition	User session is created and the user is logged into their dashboard.

Use Case: Manage Timetable

Field	Detail
Use Case Name	Manage Timetable
Actor	Administrator / Academic In-Charge
Preconditions	User must be logged in with sufficient privileges.
Main Flow	1. User selects Class and Period. 2. User selects Subject, Teacher, Room, Section, and Days. 3. System checks for scheduling conflicts. 4. If no conflict, the entry is saved and grid is refreshed.
Alternate Flow	If a conflict is detected (same teacher, overlapping day/period, different class), the system rejects the entry and displays a conflict message.
Postcondition	Timetable entry is added/updated in the database.

Use Case: View Personal Timetable

Field	Detail
Use Case Name	View Personal Timetable
Actor	Teacher / Student

Field	**Detail**
Preconditions	User must be logged in.
Main Flow	1. User navigates to "My Timetable". 2. System filters the master timetable to only the entries relevant to the logged-in user. 3. System displays the filtered grid.
Alternate Flow	If no entries exist, system displays "No classes assigned yet."
Postcondition	User views their personal schedule on screen.

CHAPTER 3:

DESIGN AND ANALYSIS

3.1 System Architecture

3-Tier Architecture of CTT System

The College Timetable Tracker (CTT) is built on a classic **3-tier architecture**, which separates the system into three distinct layers: Presentation, Application, and Data. Each tier has a specific role and communicates with the adjacent tier in a structured way.

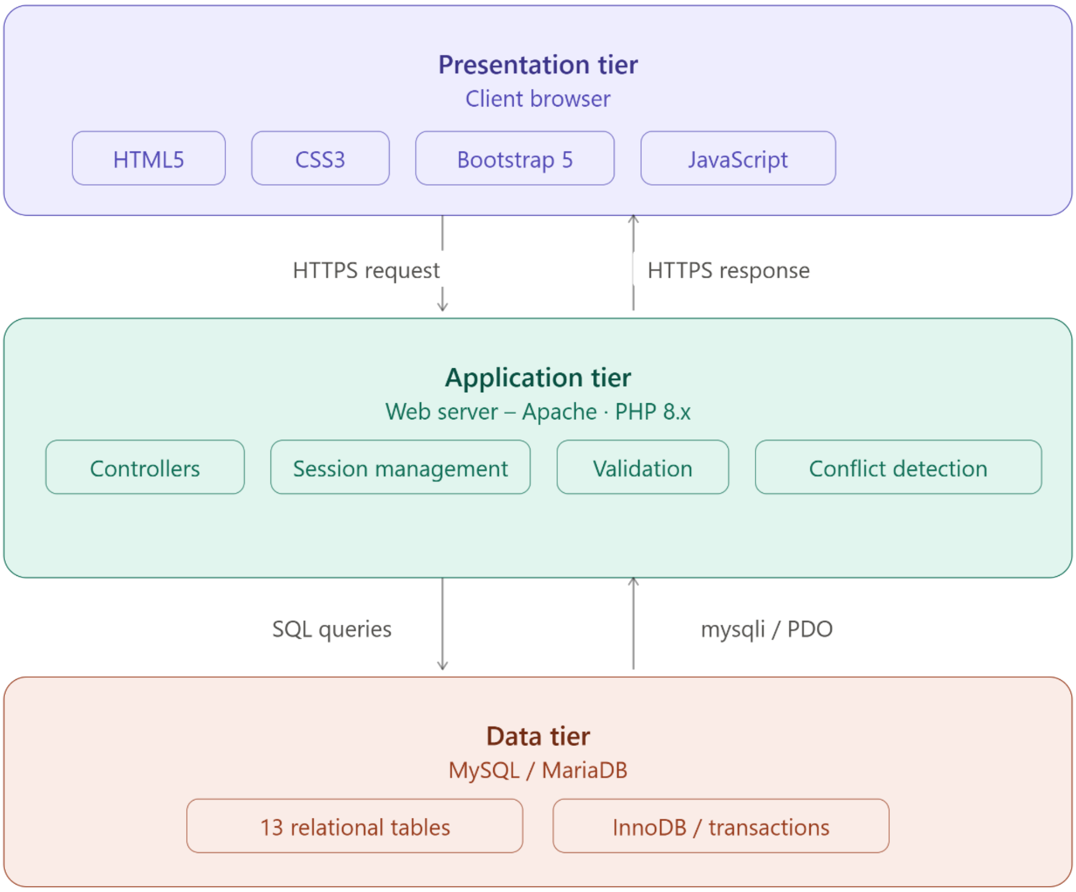


Figure 3.1: System Architecture (3-Tier)

Communication Flow

- User browser → HTTPS request → Application tier
- Application tier → SQL queries → Database
- Database → mysqli/PDO → Application tier
- Application tier → HTTPS response → User browser

This architecture ensures a clean separation of concerns, making the system more secure, maintainable, and scalable.

3.2 Data Flow Diagram

1. Level 0 DFD

The CTT system acts as the central process that receives inputs from and provides outputs to all users. There are four main actors involved:

- **Administrator** — The admin manages user accounts, timetables, and generates reports. The admin has the highest level of access in the system.
- **Academic In-Charge** — This role is responsible for managing teachers, students, and timetable assignments within their department.
- **Teacher** — The teacher interacts with the system to view their personal timetable, showing which classes they are assigned to teach.
- **Student** — The student uses the system to view their personal class timetable based on their enrolled program and section.

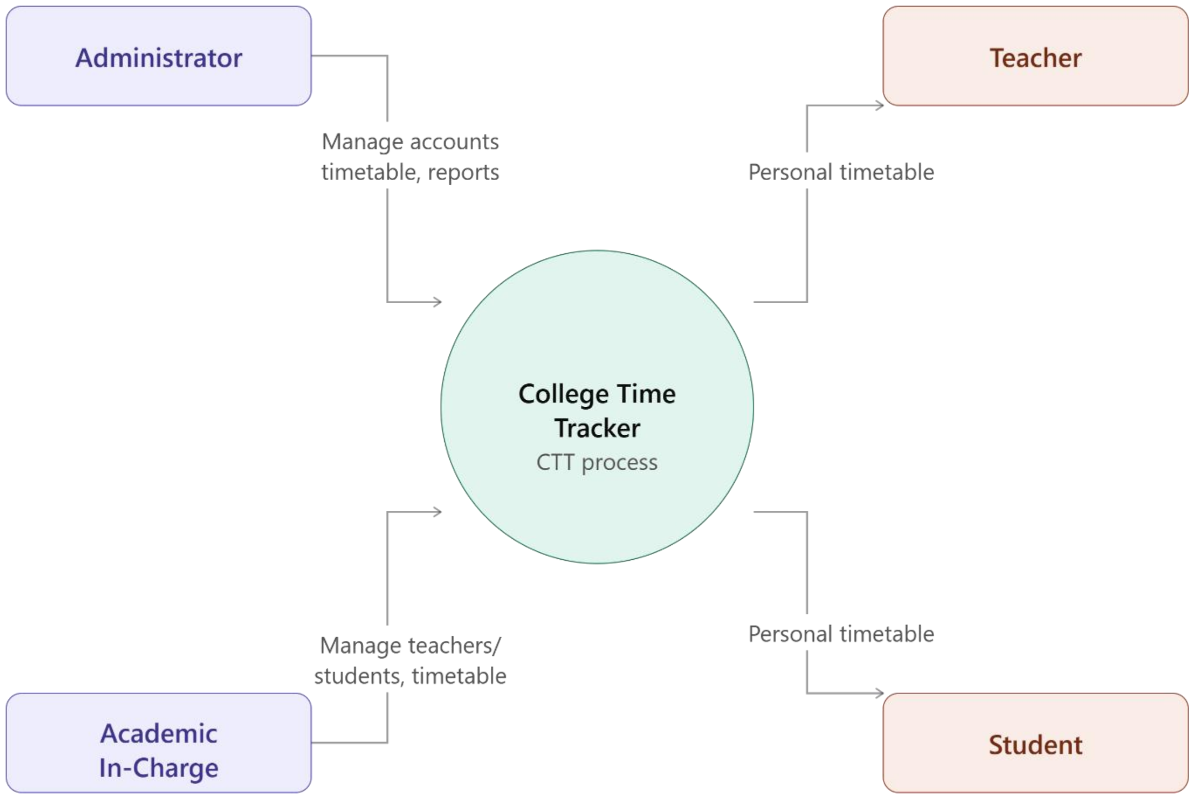


Figure 3.2: Context Level DFD (Level 0)

2. Level 1 DFD

The internal processes of the CTT system into five main functions, showing how data flows between actors, processes, and data stores.

- **Process 1.0 – Manage Users:** The Administrator sends account management requests to this process, which writes user data to the **D1 Users** data store. This includes creating and managing accounts for teachers, students, and in-charges.
- **Process 2.0 – Manage Academic Entities:** The Academic In-Charge manages entities such as programs, classes, subjects, and rooms. This process writes all academic data to the **D2 data store** (Programs / Classes / Subjects / Rooms).
- **Process 3.0 – Build Timetable + Conflict Check:** Admin and In-Charge provide timetable input to this process. It writes timetable data to the **D3 Timetable data** store and also performs automatic conflict detection to prevent scheduling errors.
- **Process 4.0 – View Personal Timetable:** Teachers and Students send view requests to this process, which reads data from the **D3 Timetable data** store and returns the relevant personal timetable to each user.
- **Process 5.0 – Generate Reports:** This process reads data from both D2 and D3 data stores and generates reports for Admin and In-Charge, providing a complete overview of academic and timetable information.

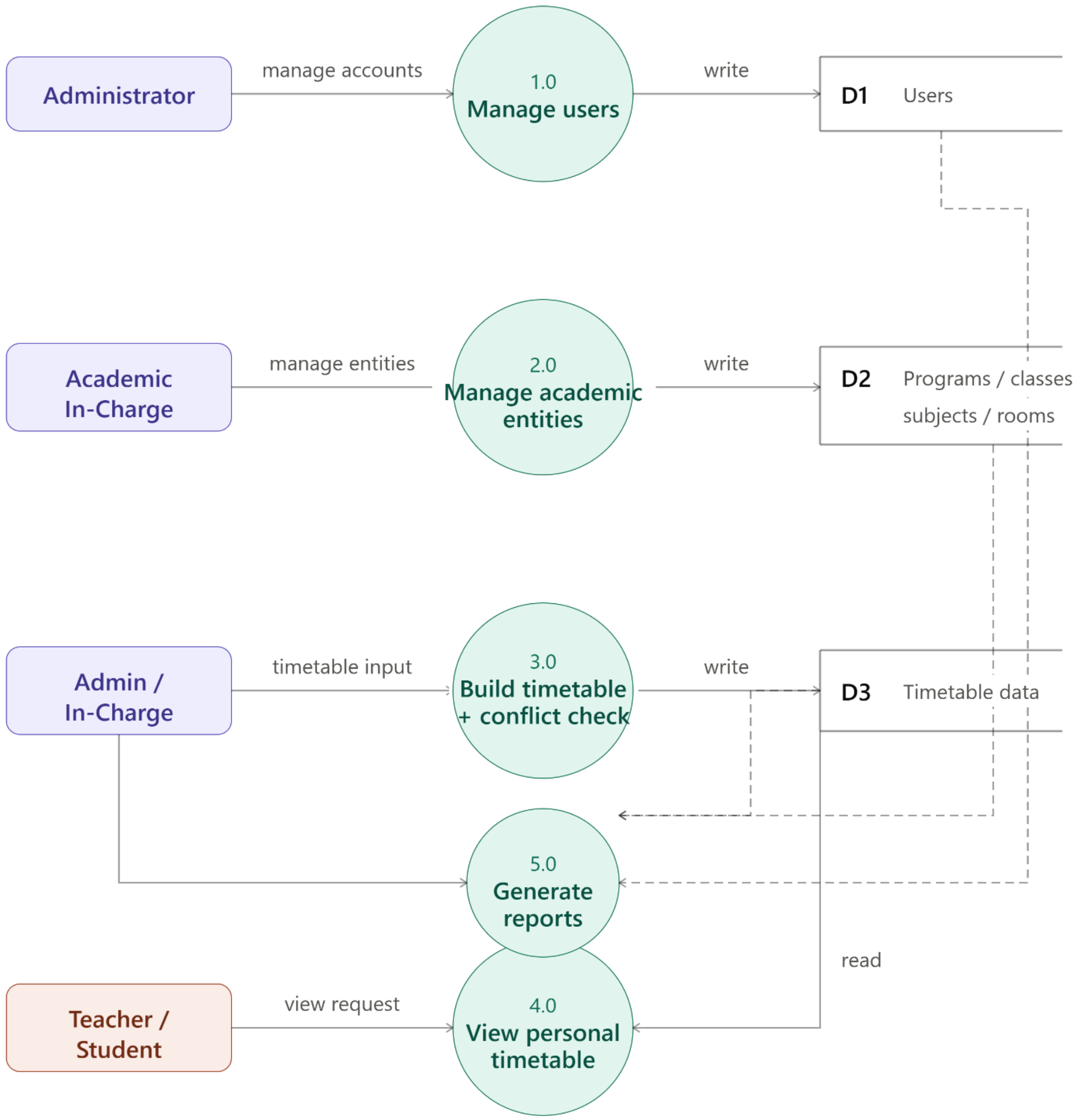


Figure 3.3: Level 1 DFD

3.3 Entity Relationship Diagram

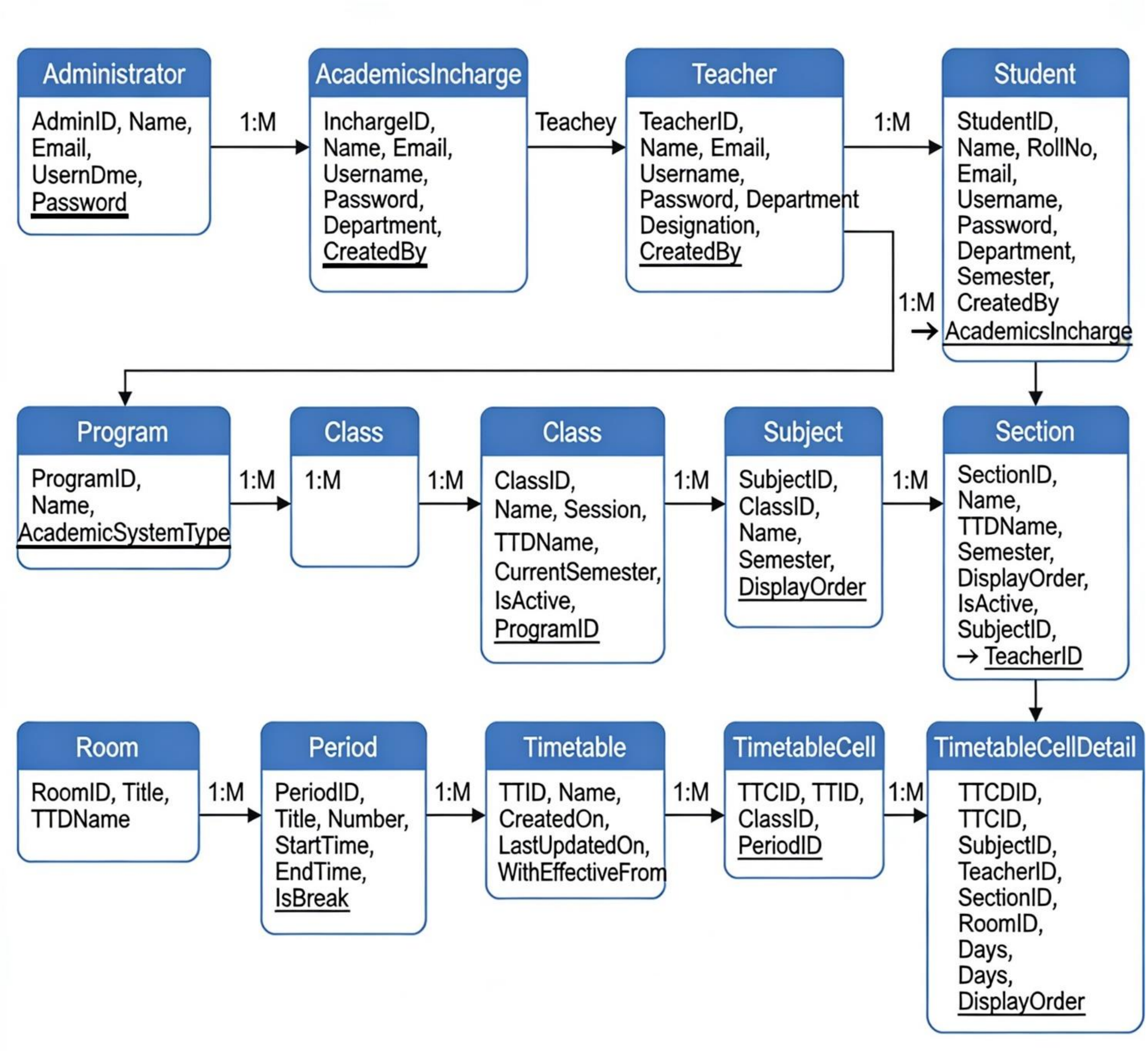


Figure 3.4: Entity Relationship Diagram

Key Entities, Attributes, Primary Keys (underlined), and Foreign Keys:

Entity	Attributes	PK	FK	Relationship
Administrator	AdminID, Name, Email, Username, Password	AdminID	—	1:M → AcademicsIncharge

Final Year Project

Entity	Attributes	PK	FK	Relationship
AcademicsIncharge	InchargeID, Name, Email, Username, Password, Department, CreatedBy	InchargeID	CreatedBy → Administrator	1:M → Teacher, Student
Teacher	TeacherID, Name, Email, Username, Password, Department, Designation, CreatedBy	TeacherID	CreatedBy → AcademicsIncharge	1:M → Section
Student	StudentID, Name, RollNo, Email, Username, Password, Department, Semester, CreatedBy	StudentID	CreatedBy → AcademicsIncharge	—
Program	ProgramID, Name, AcademicSystemType	ProgramID	—	1:M → Class
Class	ClassID, Name, Session, TTDName, CurrentSemester, IsActive, ProgramID	ClassID	ProgramID → Program	1:M → Subject
Subject	SubjectID, ClassID, Name, Semester, DisplayOrder	SubjectID	ClassID → Class	1:M → Section
Section	SectionID, Name, TTDName, Semester, DisplayOrder,	SectionID	SubjectID → Subject, TeacherID → Teacher	M:1

Entity	Attributes	PK	FK	Relationship
	IsActive, SubjectID, TeacherID			
Room	RoomID, Title, TTDName	RoomID	—	1:M → TimetableCellDetail
Period	PeriodID, Title, Number, StartTime, EndTime, IsBreak	PeriodID	—	1:M → TimetableCell
Timetable	TTID, Name, CreatedOn, LastUpdatedOn, WithEffectiveFrom	TTID	—	1:M → TimetableCell
TimetableCell	TTCID, TTID, ClassID, PeriodID	TTCID	TTID, ClassID, PeriodID	1:M → TimetableCellDetail
TimetableCellDetail	TTCDID, TTCID, SubjectID, TeacherID, SectionID, RoomID, Days, DisplayOrder	TTCDID	TTCID, SubjectID, TeacherID, SectionID, RoomID	M:1

Cardinalities: Administrator (1) — (M) AcademicsIncharge; AcademicsIncharge (1) — (M) Teacher/Student; Program (1) — (M) Class (1) — (M) Subject (1) — (M) Section; TimetableCell (1) — (M) TimetableCellDetail.

3.4 UML Diagrams

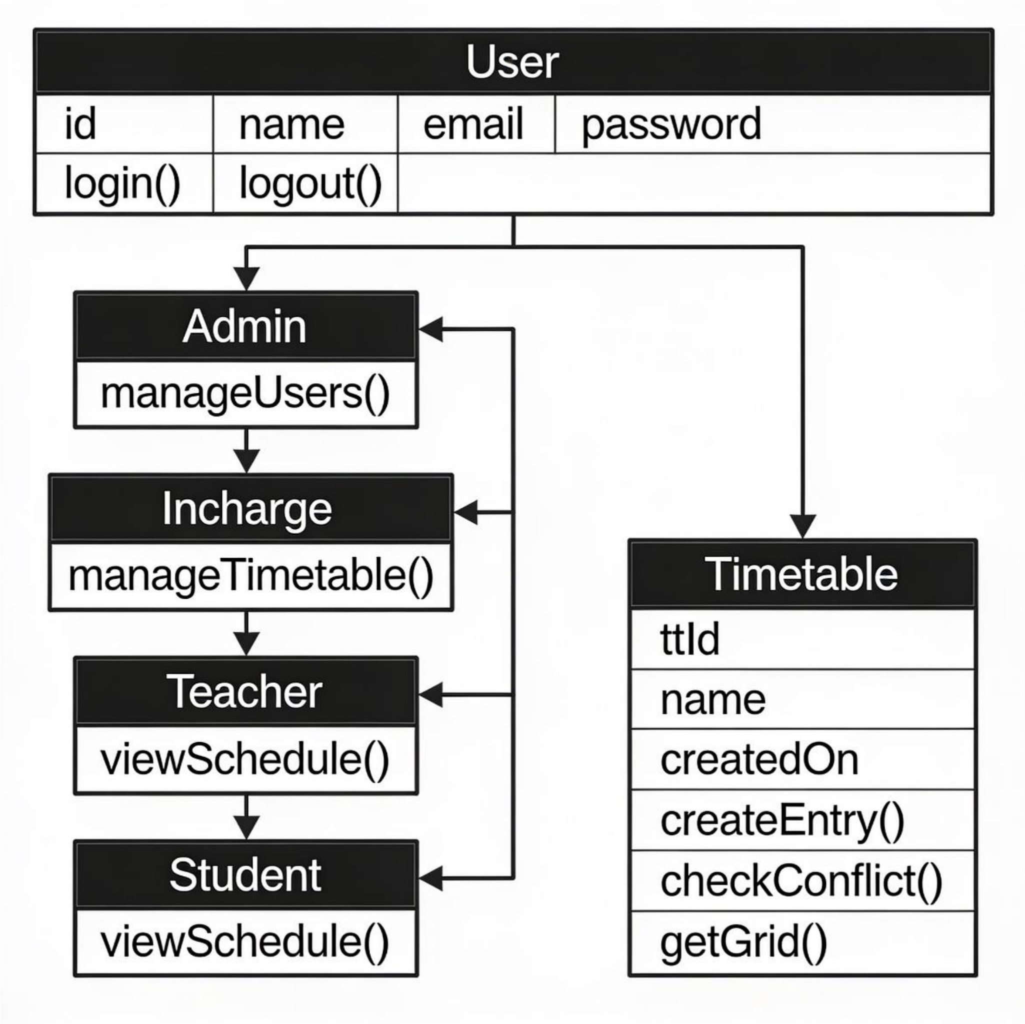


Figure 3.5: Class Diagram (Key Classes)

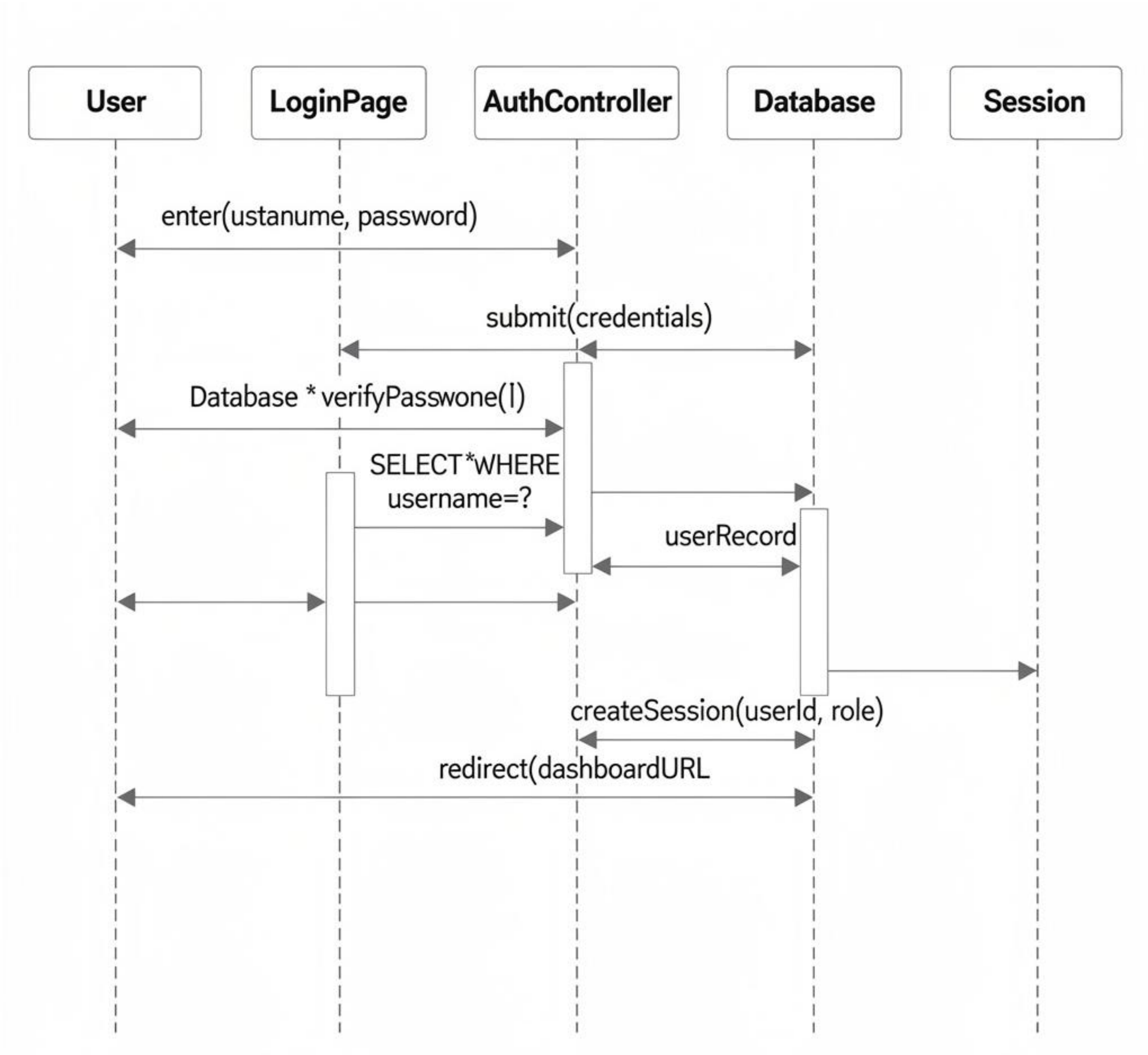


Figure 3.6: Sequence Diagram — Login

Description

User → LoginPage: enter(username, password)

LoginPage → AuthController: submit(credentials)

AuthController → Database: SELECT * WHERE username=?

Database → AuthController: userRecord

AuthController → AuthController: verifyPassword()

AuthController → Session: createSession(userId, role)

AuthController → LoginPage: redirect(dashboardURL)

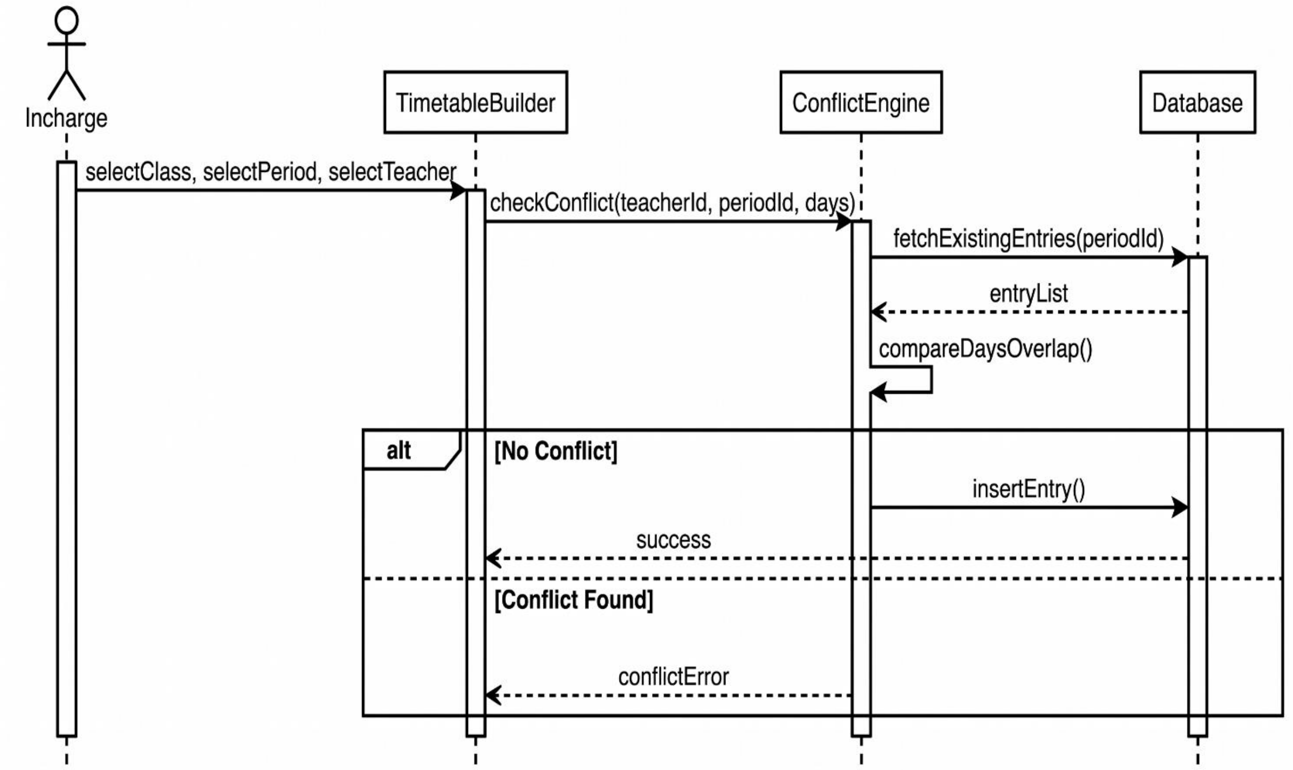


Figure 3.7: Sequence Diagram — Create Timetable Entry

Description

Incharge → TimetableBuilder: selectClass, selectPeriod, selectTeacher

TimetableBuilder → ConflictEngine: checkConflict(teacherId, periodId, days)

ConflictEngine → Database: fetchExistingEntries(periodId)

Database → ConflictEngine: entryList

ConflictEngine → ConflictEngine: compareDaysOverlap()

If No Conflict: ConflictEngine → Database: insertEntry(); Database → TimetableBuilder: success

If Conflict Found: ConflictEngine → TimetableBuilder: conflictError

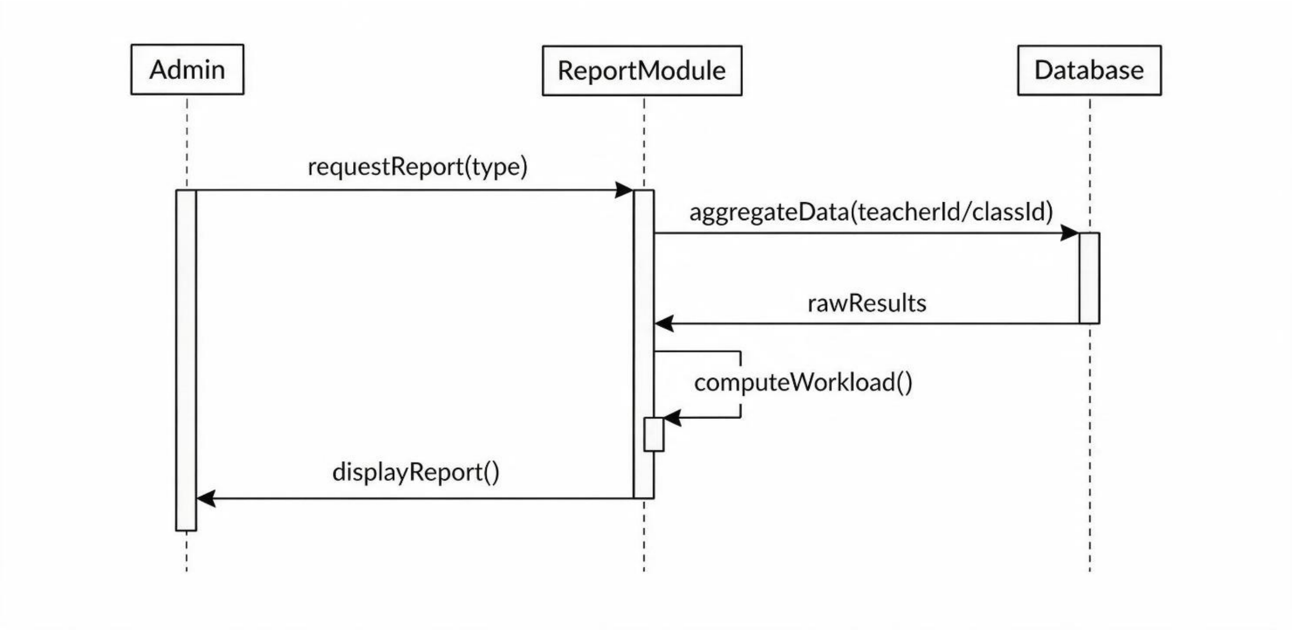


Figure 3.8: Sequence Diagram — Generate Report

Description

Admin → ReportModule: requestReport(type)

ReportModule → Database: aggregateData(teacherId/classId)

Database → ReportModule: rawResults

ReportModule → ReportModule: computeWorkload()

ReportModule → Admin: displayReport()

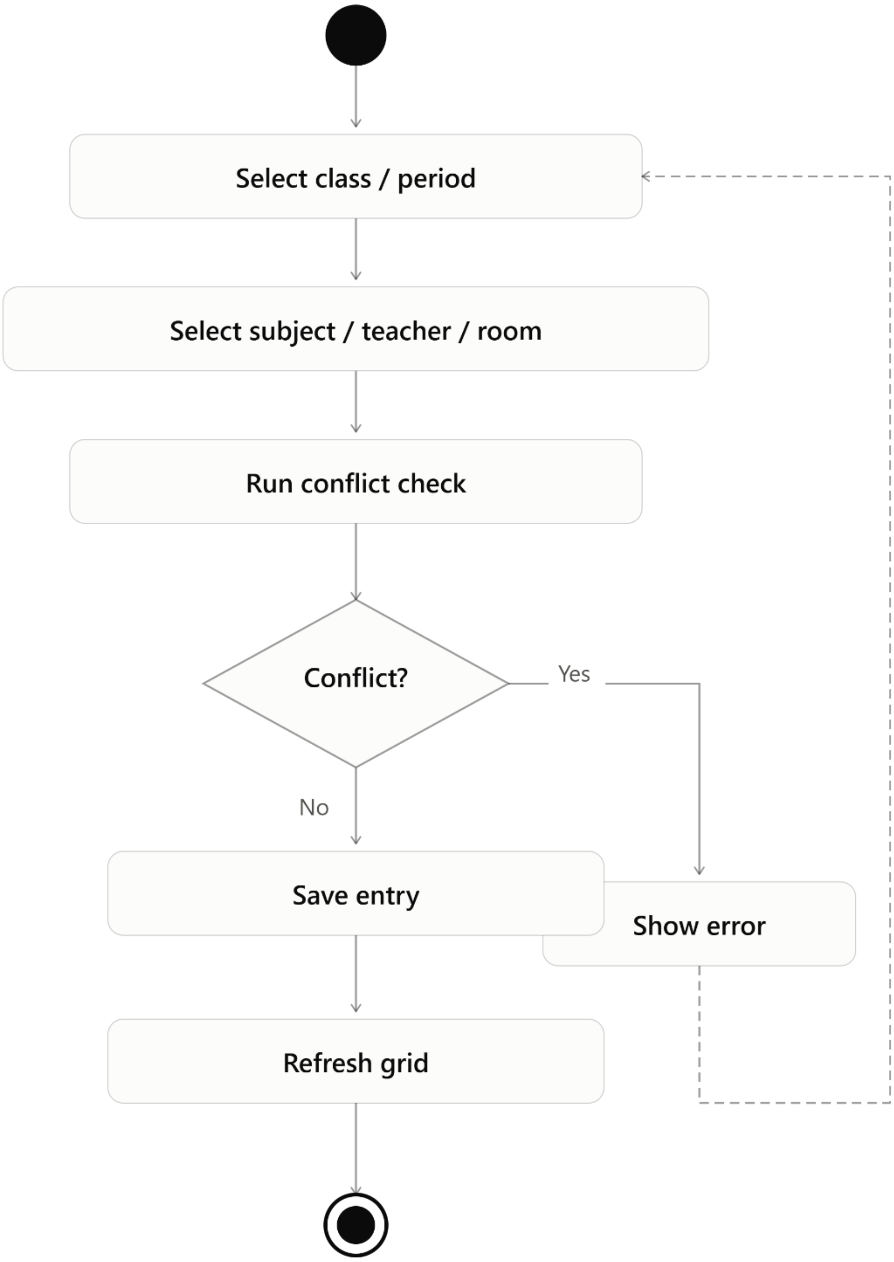


Figure 3.9: Activity Diagram — Timetable Creation

3.5 Database Design

Table 3.1: Database Tables Description

Table Name	Field Name	Data Type	Description
administrator	AdminID	INT (PK)	Unique admin identifier

Table Name	Field Name	Data Type	Description
administrator	Username	VARCHAR(50)	Login username
administrator	Password	VARCHAR(255)	Hashed password
academicsincharge	InchargeID	INT (PK)	Unique in-charge identifier
academicsincharge	CreatedBy	INT (FK)	References Administrator
teacher	TeacherID	INT (PK)	Unique teacher identifier
teacher	Designation	VARCHAR(100)	Teacher's academic designation
student	StudentID	INT (PK)	Unique student identifier
student	RollNo	VARCHAR(50)	Student roll number
timetable	TTID	INT (PK)	Timetable record identifier
timetablecell	TTCID	INT (PK)	Grid cell identifier
timetablecell	ClassID	INT (FK)	References Class
timetablecell	PeriodID	INT (FK)	References Period
timetablecelldetail	TTCDID	INT (PK)	Cell-detail identifier
timetablecelldetail	Days	VARCHAR(50)	Days pattern (e.g., MO-TU-WE)

Administrator Table 3.1

Field	Type	Description

AdminID PK	int(11)	Unique administrator ID
Name	varchar(100)	Full name
Email	varchar(100)	Unique email
Username	varchar(50)	Login username
Password	varchar(255)	Encrypted password

AcademicsIncharge Table 3.2

Field	**Type**	**Description**
InchargeID PK	int(11)	Unique ID
Name	varchar(100)	Full name
Email	varchar(100)	Unique email
Username	varchar(50)	Login username
Password	varchar(255)	Encrypted password
Department	varchar(100)	Assigned department
CreatedBy FK	int(11)	References AdminID

Program Table 3.3

Field	**Type**	**Description**
ProgramID PK	int(11)	Unique program ID
Name	varchar(100)	Program name (BSCS, BSIT…)
AcademicSystemType	varchar(50)	Semester / annual

Class Table 3.4

Field	**Type**	**Description**

ClassID PK	int(11)	Unique class ID
Name	varchar(100)	Class name
Session	varchar(50)	Academic session
TTDName	varchar(50)	Timetable display name
CurrentSemester	varchar(50)	Current semester
IsActive	tinyint(1)	Active / inactive status
ProgramID FK	int(11)	References Program

Teacher Table 3.5

Field	**Type**	**Description**
TeacherID PK	int(11)	Unique teacher ID
Name	varchar(100)	Teacher name
Email	varchar(100)	Unique email
Username	varchar(50)	Login username
Password	varchar(255)	Encrypted password
Department	varchar(100)	Department name
Designation	varchar(100)	Academic designation
CreatedBy FK	int(11)	References InchargeID
MaxHoursPerDay	int(11)	Max teaching hours/day

Student Table 3.6

Field	**Type**	**Description**
StudentID PK	int(11)	Unique student ID

Name	varchar(100)	Student name
RollNo	varchar(50)	Unique roll number
Email	varchar(100)	Unique email
Username	varchar(50)	Login username
Password	varchar(255)	Encrypted password
Department	varchar(100)	Department name
Semester	varchar(50)	Current semester
CreatedBy FK	int(11)	References InchargeID

Subjects Table 3.7

Field	Type	Description
SubjectID PK	int(11)	Unique subject ID
ClassID FK	int(11)	References Class
Name	varchar(100)	Subject name
Semester	varchar(50)	Assigned semester
DisplayOrder	int(11)	Display sequence
WeeklyHours	int(11)	Weekly lecture hours
IsLab	tinyint(1)	Lab subject indicator

Section Table 3.8

Field	Type	Description
SectionID PK	int(11)	Unique section ID
Name	varchar(100)	Section name

TTDName	varchar(50)	Timetable display name
Semester	varchar(50)	Semester
DisplayOrder	int(11)	Display order
IsActive	tinyint(1)	Active status
SubjectID FK	int(11)	References Subject
TeacherID FK	int(11)	References Teacher

Room Table 3.9

Field	**Type**	**Description**
RoomID PK	int(11)	Unique room ID
Title	varchar(50)	Room name / number
TTDName	varchar(50)	Timetable display name
IsLab	tinyint(1)	Lab room indicator

Period Table 3.10

Field	**Type**	**Description**
PeriodID PK	int(11)	Unique period ID
Title	varchar(50)	Period title
Number	int(11)	Period number
StartTime	time	Start time

EndTime	time	End time
FridayStartTime	time	Friday start time
FridayEndTime	time	Friday end time
IsZeroPeriod	tinyint(1)	Zero period indicator
IsBreak	tinyint(1)	Break indicator
DisplayOrder	int(11)	Display order

TimeTable Table 3.11

Field	**Type**	**Description**
TTID PK	int(11)	Timetable record ID
Name	varchar(100)	Timetable name
CreatedOn	date	Creation date
LastUpdatedOn	date	Last updated date
WithEffectiveFrom	date	Effective from date

TimetableCell Table 3.12

Field	**Type**	**Description**
TTCID PK	int(11)	Grid cell ID
TTID FK	int(11)	References TimeTable

ClassID FK	int(11)	References Class
PeriodID FK	int(11)	References Period

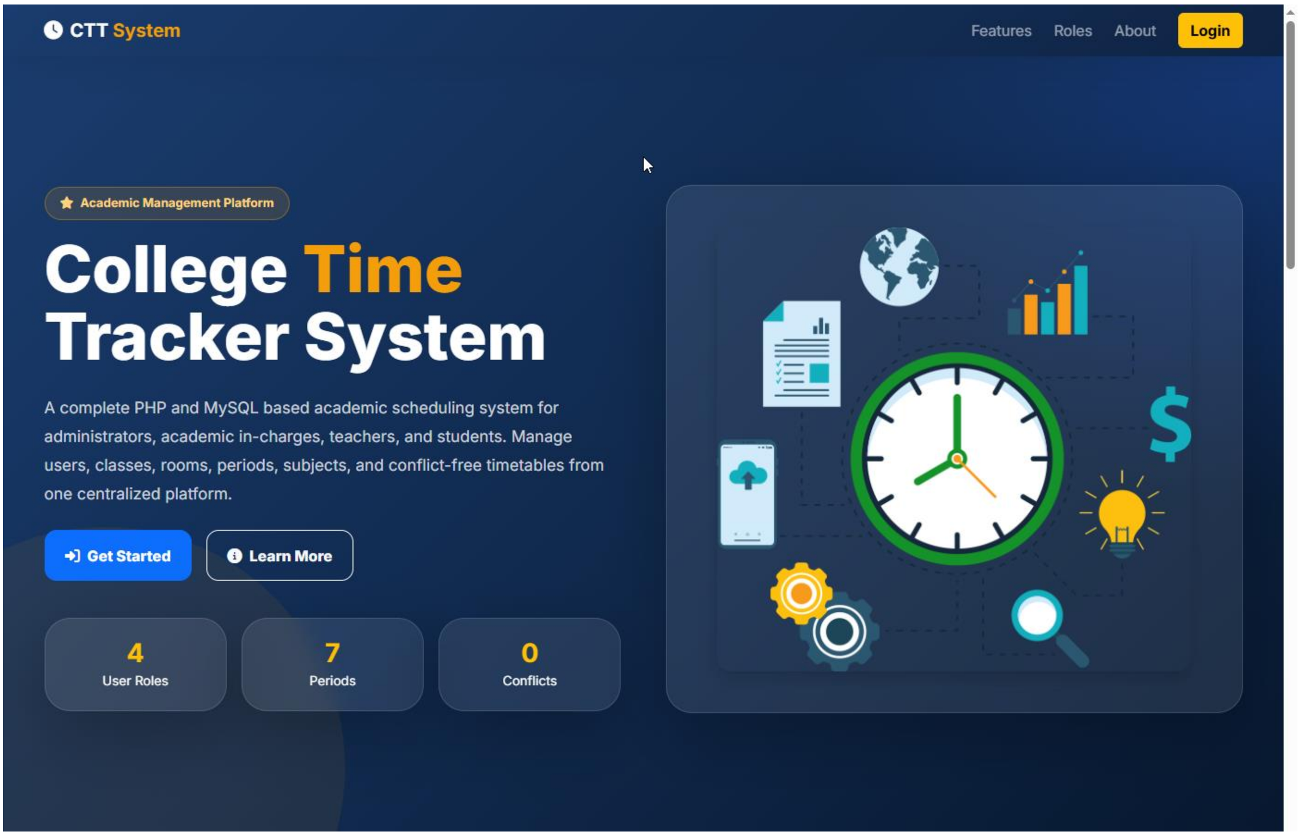
TimetableCellDetail Table 3.13

Field	**Type**	**Description**
TTCDID PK	int(11)	Cell-detail ID
TTCID FK	int(11)	References TimetableCell
SubjectID FK	int(11)	References Subject
TeacherID FK	int(11)	References Teacher
SectionID FK	int(11)	References Section
RoomID FK	int(11)	References Room
Days	varchar(20)	Day pattern (e.g. MO-TU-WE)
DisplayOrder	int(11)	Display sequence

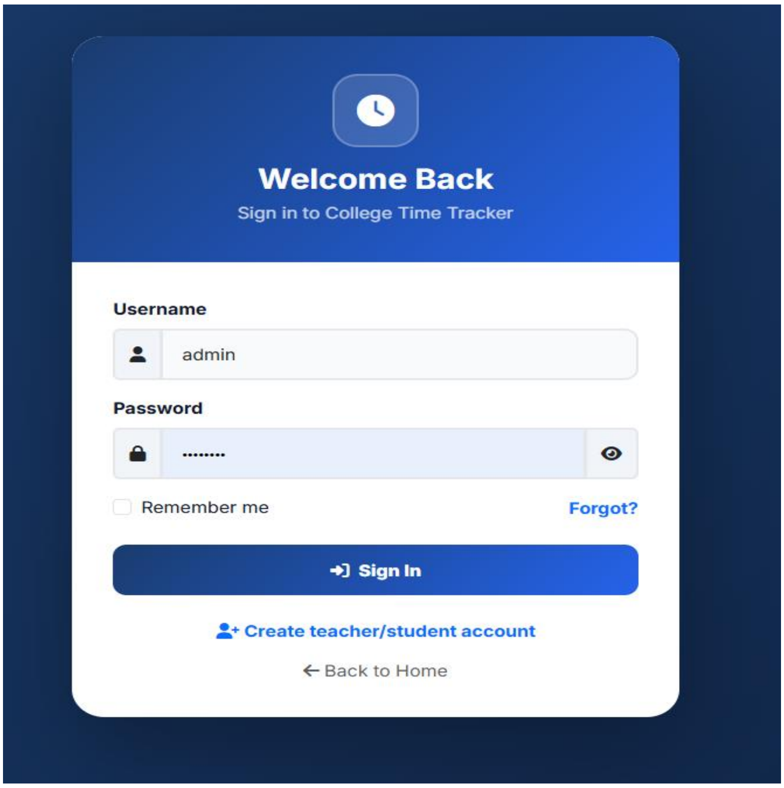
3.6 User Interface Design

- Wireframes: Home page → Login page → Role Dashboard → Timetable Grid → Management forms.
- Mockups: Centered card-based login form; sidebar navigation dashboard layout; grid-based timetable view color-coded by class.
- Navigation Structure: Top-level: Home → Login/Register. After login: Sidebar → Dashboard, User Management, Academic Entities, Timetable Builder, Reports.

Home page

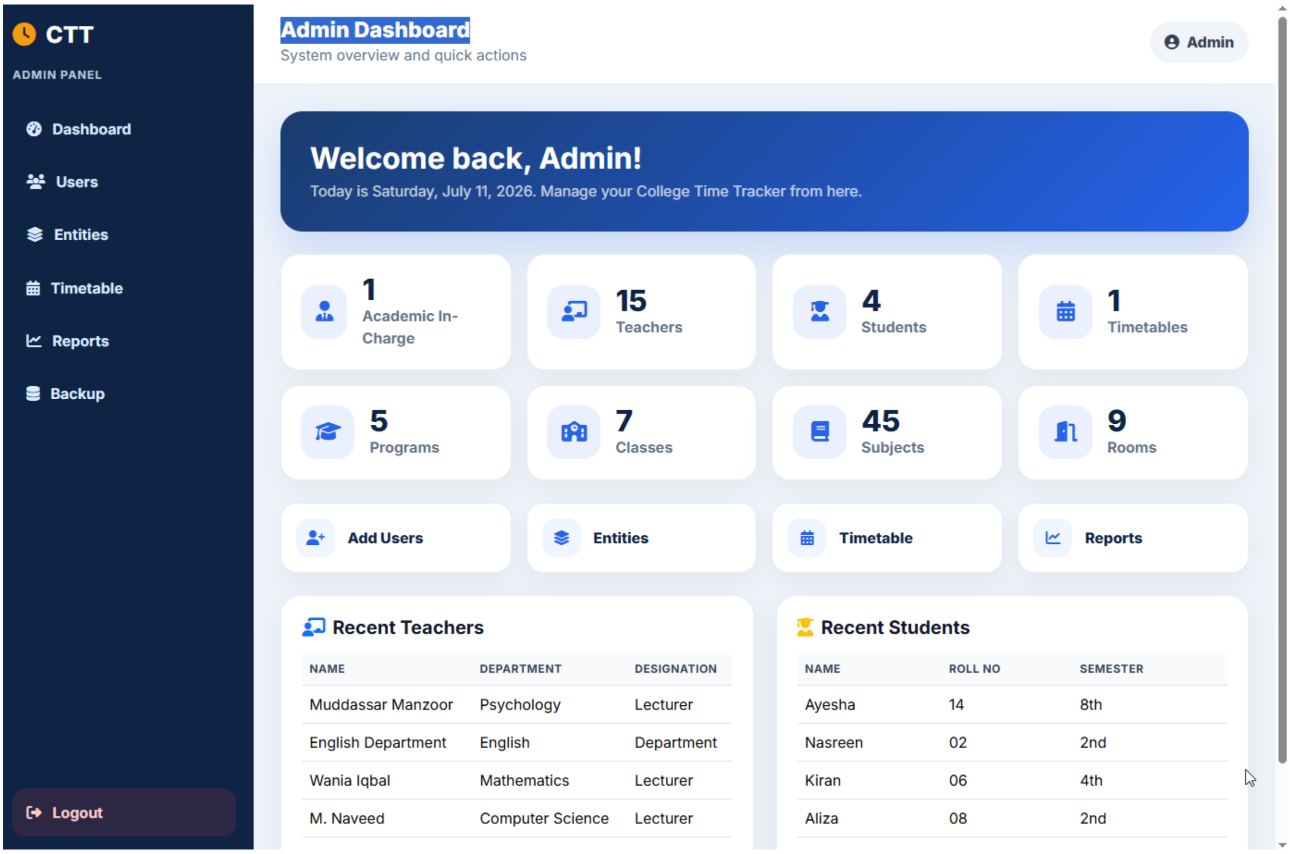


Login page

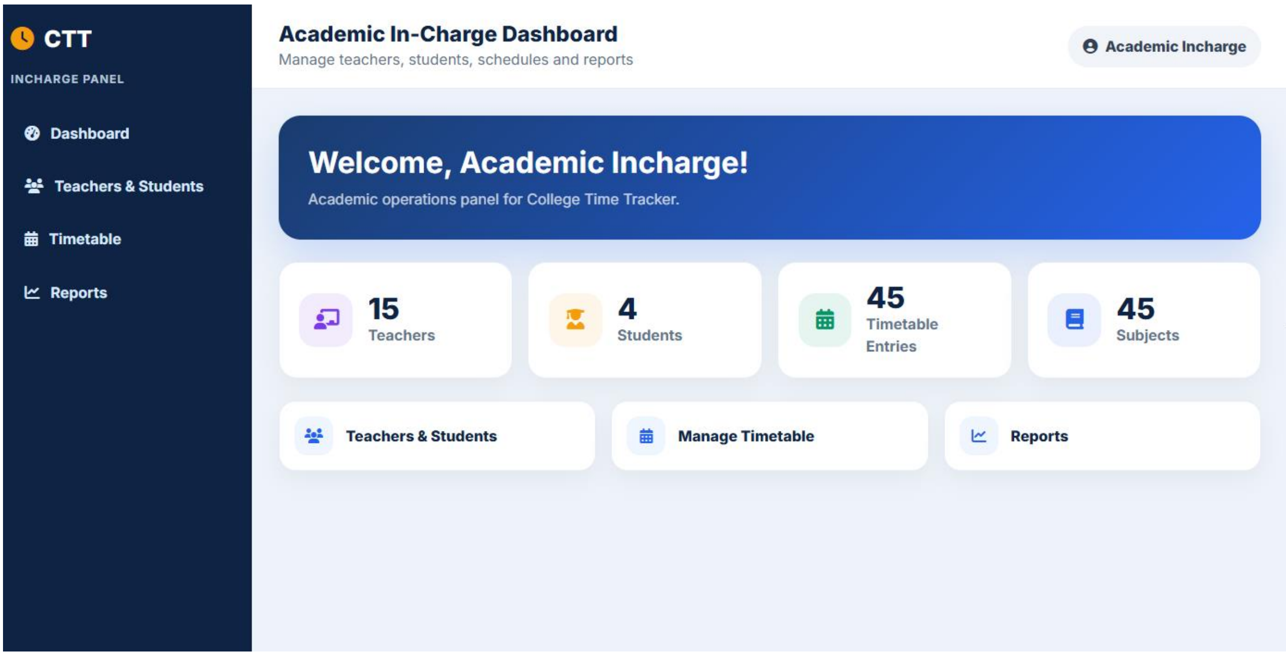


Role Dashboard

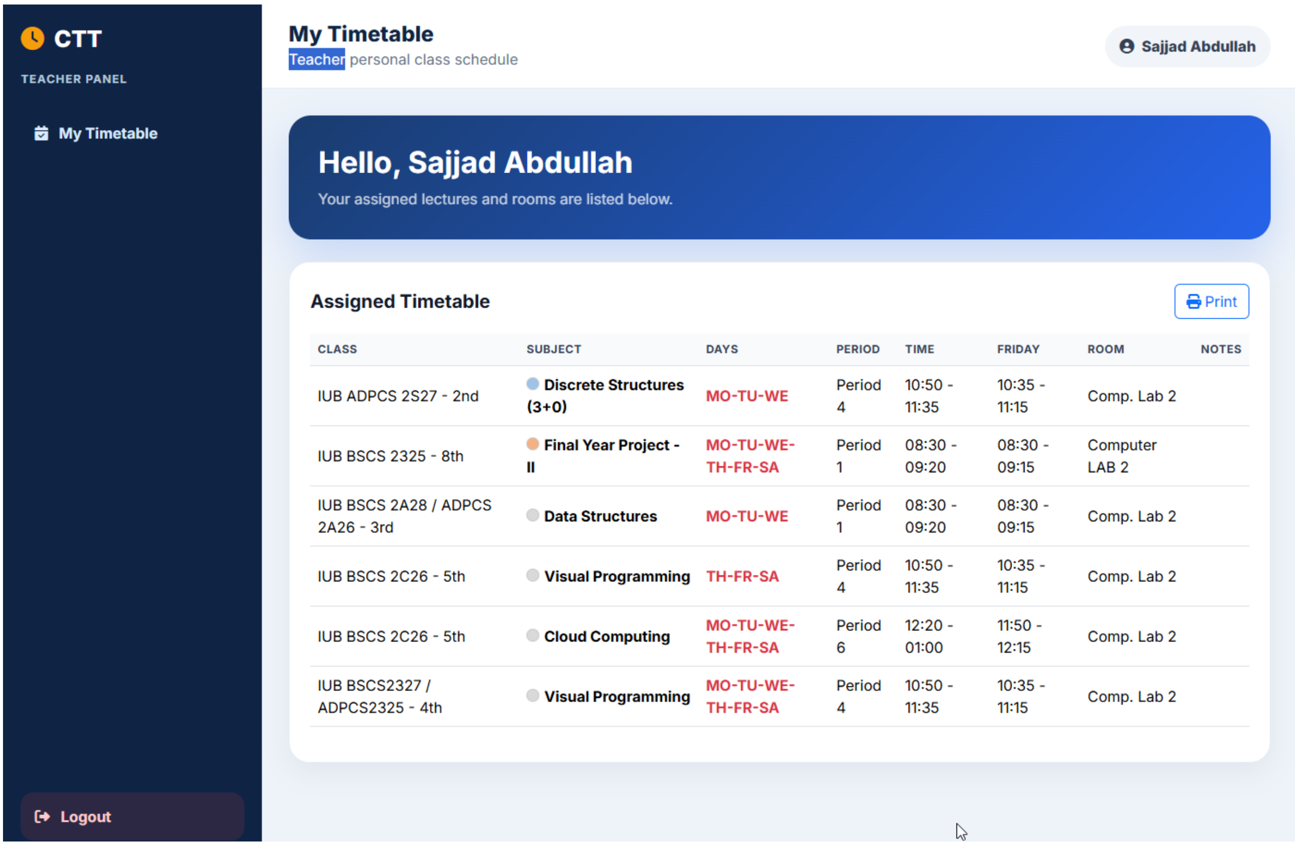
1. Admin Dashboard



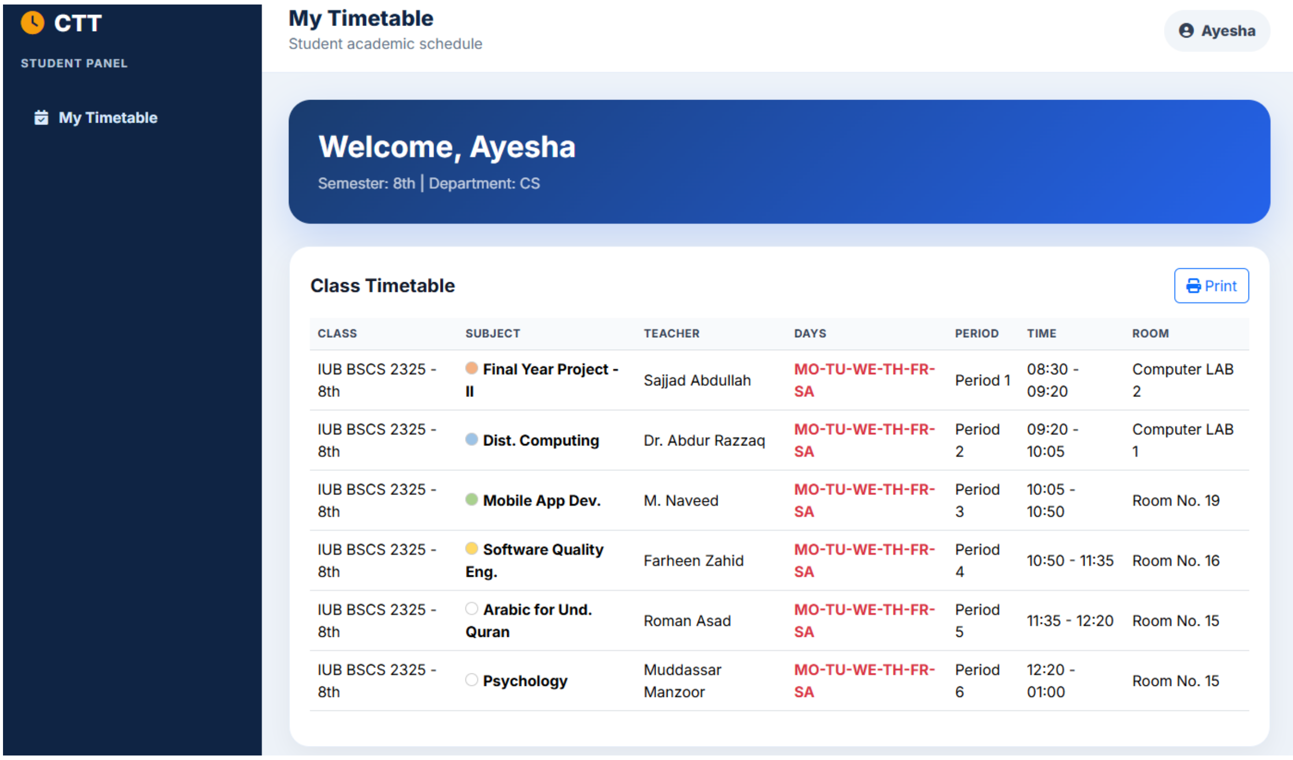
2. Academic In-Charge Dashboard



3. Teacher Dashboard



4. Student Dashboard



CHAPTER 4:

DEVELOPMENT

4.1 Development Environment

Item	Detail
Operating System	Windows 10/11
IDE	Visual Studio Code
Local Server	XAMPP (Apache + MySQL + PHP 8.x)
Version Control	Git / GitHub
Browser Testing	Google Chrome, Mozilla Firefox

4.2 System Modules

4.2.1 Authentication Module

- Purpose: Validate user credentials and establish a role-based session.
- Input: Username, Password.
- Output: Authenticated session redirecting to the correct dashboard.

4.2.2 Admin Module

- Purpose: Allow the Administrator to manage Academic In-Charge accounts and system-wide settings.
- Input: In-Charge details (Name, Email, Department).
- Output: Updated user list with confirmation message.

4.2.3 Timetable Module

- Purpose: Build and manage the master timetable grid with conflict prevention.
- Input: Class, Period, Subject, Teacher, Room, Section, Days.
- Output: Updated timetable grid, or a conflict error message.

4.2.4 Reporting Module

- Purpose: Generate teacher workload and class-load summaries.
- Input: Date range / filter criteria.
- Output: Tabular report, exportable to CSV.

4.3 Key Implementation Details

Conflict Detection Algorithm: Before inserting a new timetable entry, the system retrieves all existing entries for the same Period, filters those with a different Class, and checks whether the assigned Teacher matches and whether the Days values share any overlapping day token (e.g., “MO”). If both conditions are true, the entry is rejected.

4.3.1 Description of the Conflict Check Logic

This subsection explains, in narrative form and without displaying a code listing, exactly how the conflict-checking routine referred to in the system as hasConflict() behaves. The purpose of this explanation is to describe the same logic that was originally implemented as a PHP function, but expressed entirely as descriptive text so that a reader unfamiliar with PHP syntax can still understand precisely what the routine checks, in what order, and why each step is necessary. No behavior described below has been altered from the original implementation; only the presentation has changed from a code block to a detailed written explanation.

Purpose of the routine. The routine exists to answer a single yes/no question before any timetable entry is written to the database: “Is the teacher who is about to be assigned to this class, during this period, on these days, already teaching a different class at an overlapping time?” If the answer is yes, the routine reports that a conflict exists and the calling code must refuse to save the new entry. If the answer is no, the routine reports that no conflict exists and the calling code is free to proceed with the insert or update operation.

Inputs accepted by the routine. The routine accepts four pieces of information every time it is invoked, together with an active database connection so that it can query the underlying tables:

The identifier of the Teacher who is being assigned to the new entry (referred to as the Teacher ID).

The identifier of the Period in which the new entry is being placed (referred to as the Period ID), since a conflict can only occur when two entries fall in the same period slot.

The Days value associated with the new entry — a short text pattern such as “MO-TU-WE” that lists, using two-letter abbreviations separated by hyphens, which days of the week the class meets during that period.

The identifier of the Class that is currently being edited (referred to as the excluded Class ID), which is used so that the routine does not mistakenly flag the entry currently being edited as a conflict with itself.

Step 1 — Breaking the Days pattern into individual day tokens. The first action the routine takes is to split the Days text supplied for the new entry wherever a hyphen character appears. For example, the pattern “MO-TU-WE” is broken into three separate tokens: “MO”, “TU”, and “WE”. This step is necessary because a class does not always meet on a single day; many classes recur on two or three days within the same period slot, and the conflict check must be able to compare each individual day separately rather than treating the whole pattern as one indivisible string.

Step 2 — Retrieving all existing entries that could possibly conflict. The routine next queries the database for every row already stored in the timetable-cell-detail table that is linked, through the timetable-cell table, to the same Period ID that was supplied as input, and that was created by the same Teacher ID that was supplied as input. Critically, the query also excludes any existing rows that belong to the same Class ID that is currently being edited, because it would be incorrect to flag an existing entry as a conflict with itself when a user is simply updating or re-saving an entry that already belongs to that class. In effect, this step gathers a candidate list containing only those existing timetable rows where the same teacher is already committed to teaching a different class during the same period slot.

Step 3 — Comparing day patterns for overlap. For every row returned by the database query in Step 2, the routine repeats the same splitting operation described in Step 1, breaking that existing row's stored Days value into its own list of individual day tokens. It then compares the list of day tokens belonging to the new entry against the list of day tokens belonging to each existing row, looking specifically for any day token that appears in both lists. This comparison is what technically corresponds to computing the intersection of the two token sets: if the intersection is empty, the two entries never actually meet on the same calendar day even though they share the same period slot, and therefore there is no real scheduling clash; if the intersection contains at least one shared day token, then the teacher would genuinely be required to be in two different classrooms at the same time on that day, which is precisely the situation the system must prevent.

Step 4 — Reaching a verdict. As soon as the routine finds one existing row whose day tokens overlap with the new entry's day tokens, it immediately concludes that a conflict has been found and reports this outcome back to the calling code without needing to examine any further rows, since a single genuine overlap is sufficient to reject the new entry. If the routine finishes comparing the new entry against every candidate row retrieved in Step 2 and never finds an overlapping day token, it reports that no conflict was found, and the calling code is then permitted to insert the new timetable entry into the database.

Why each safeguard matters. Restricting the database query to the same Period ID ensures the routine never wastes time comparing entries that occur at completely different times of day, which could never conflict regardless of which days they fall on. Restricting the query to the same Teacher ID ensures the routine only raises an alarm when the actual constraint the college cares about — a single teacher physically being in two places at once — is genuinely at risk of being violated; two different teachers can freely be scheduled in the same period without any conflict. Excluding the Class ID currently being edited prevents the system from generating a false-positive conflict against the very entry the user is in the process of saving or updating. Finally, comparing day tokens individually, rather than comparing the full Days string for exact equality, correctly handles the common real-world case where two classes share only some, but not all, of their meeting days within the same period.

Outcome for the user. When the routine reports that a conflict exists, the Timetable Builder module displays a clear, human-readable error message to the Administrator or Academic In-Charge, identifying that the selected teacher is already committed elsewhere at an overlapping day and period, and the new entry is not written to the database. When the routine reports that no conflict exists, the Timetable Builder proceeds to save the new entry, and the weekly timetable grid is refreshed on screen to display the newly added Subject, Teacher, Room, Section, and Days combination in the appropriate cell.

4.3.2 Validation Logic

Server-side validation checks required fields, unique username/email constraints, and correct data types before any INSERT/UPDATE query executes.

4.3.3 Business Rule

A user account created by an Academic In-Charge cannot elevate its own role; only an Administrator may create In-Charge accounts.

4.4 APIs / Third-Party Integrations

No external third-party APIs are used in the core system. Bootstrap 5 and Font Awesome are included via CDN for UI styling and iconography only.

CHAPTER 5:

SOFTWARE TESTING

5.1 Testing Strategy

- Unit Testing: Individual PHP functions (e.g., password hashing, conflict check) tested in isolation.
- Integration Testing: Verified that the Timetable Builder correctly interacts with the Conflict Engine and Database layer.
- System Testing: Full end-to-end testing of all modules together.
- Acceptance Testing: Conducted with sample users (a mock Admin, In-Charge, Teacher, Student) to confirm the system meets requirements.

5.2 Test Plan

- Objective: Verify that all functional requirements (FR-01 to FR-17) behave correctly under normal, invalid, and boundary conditions.
- Scope: All modules — Authentication, User Management, Academic Entities, Timetable Builder, Reports.
- Approach: Manual black-box testing using a structured test case document.

5.3 Test Cases

Table 5.1: Test Cases

Test ID	Description	Input	Expected Output	Status
TC-01	Valid Admin login	admin / correct password	Redirect to Admin Dashboard	Pass
TC-02	Invalid password login	admin / wrong password	"Invalid username or password"	Pass
TC-03	Empty login fields	"" / ""	"Please fill in all fields"	Pass
TC-04	Add new Teacher	Valid teacher details	Teacher added, confirmation shown	Pass
TC-05	Add Teacher with duplicate email	Existing email	"Email already exists"	Pass

Test ID	Description	Input	Expected Output	Status
TC-06	Add Student with missing Roll No	Blank RollNo	"Roll Number is required"	Pass
TC-07	Create timetable entry (no conflict)	Unique teacher/period	Entry saved	Pass
TC-08	Create timetable entry (conflict)	Same teacher, overlapping day, different class	"Conflict detected" error	Pass
TC-09	Delete timetable entry	Valid entry ID	Entry removed from grid	Pass
TC-10	View personal timetable (Teacher)	Logged in as Teacher	Only teacher's own classes shown	Pass
TC-11	View personal timetable (Student)	Logged in as Student	Only student's class shown	Pass
TC-12	Access Admin page as Student	Student session, Admin URL	Redirected to Student dashboard	Pass
TC-13	Password field boundary (5 characters)	"abcde"	"Password must be at least 6 characters"	Pass
TC-14	Password field boundary (6 characters)	"abcdef"	Accepted	Pass
TC-15	Generate teacher workload report	Valid request	Report table displayed	Pass
TC-16	Export timetable to CSV	Click Export	CSV file downloaded	Pass
TC-17	Print timetable	Click Print	Print-friendly layout shown	Pass

Test ID	Description	Input	Expected Output	Status
TC-18	SQL Injection attempt in login	' OR '1'='1	Login rejected, no bypass	Pass
TC-19	Add Room with duplicate title	Existing room title	Room still added / warning	Pass
TC-20	Session expiry after logout	Logout action	Redirect to login, session cleared	Pass

Additional boundary/negative cases were tested for each CRUD module following the same pattern, totaling 20+ test cases.

5.4 Black Box Testing

Black-box testing was performed on all user-facing forms (login, registration, timetable builder) without inspecting internal code, focusing on input/output correctness relative to functional requirements.

5.6 Bug Reports

Bug ID	Description	Severity	Status
BUG-01	Timetable grid did not refresh immediately after adding an entry	Medium	Fixed
BUG-02	Duplicate room names were allowed	Low	Fixed
BUG-03	Session did not expire after prolonged inactivity	Medium	Fixed

CHAPTER 6:

IMPLEMENTATION

6.1 Hardware Requirements

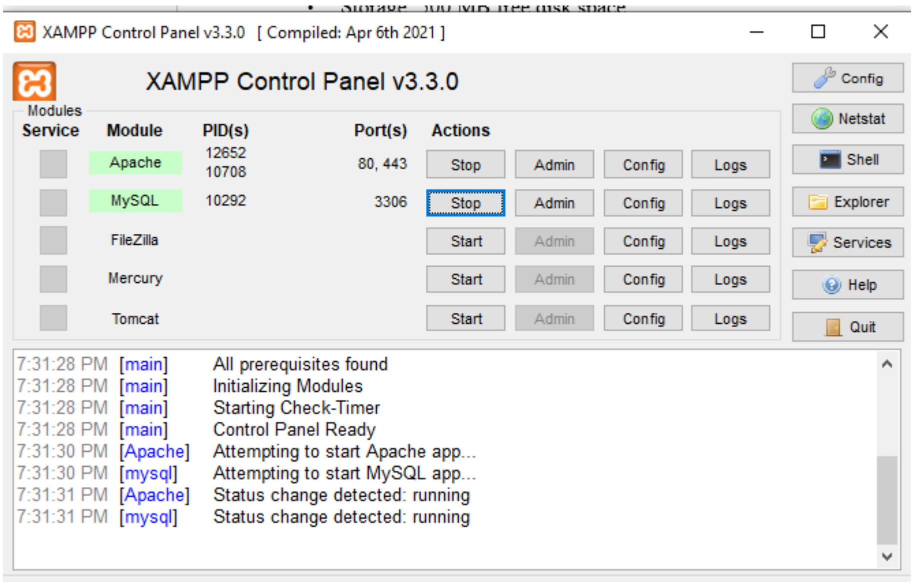
- Processor: Dual-core 2 GHz or higher
- RAM: 4 GB minimum
- Storage: 500 MB free disk space

6.2 Software Requirements

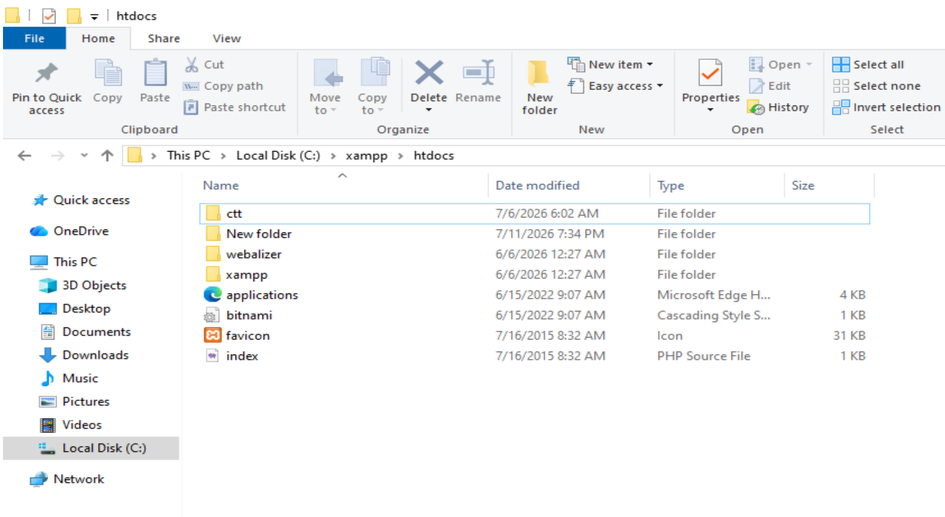
- Operating System: Windows 10/11 or Linux
- XAMPP (Apache, MySQL, PHP 8.x)
- Modern web browser (Chrome, Firefox, Edge)

6.3 Installation Procedure

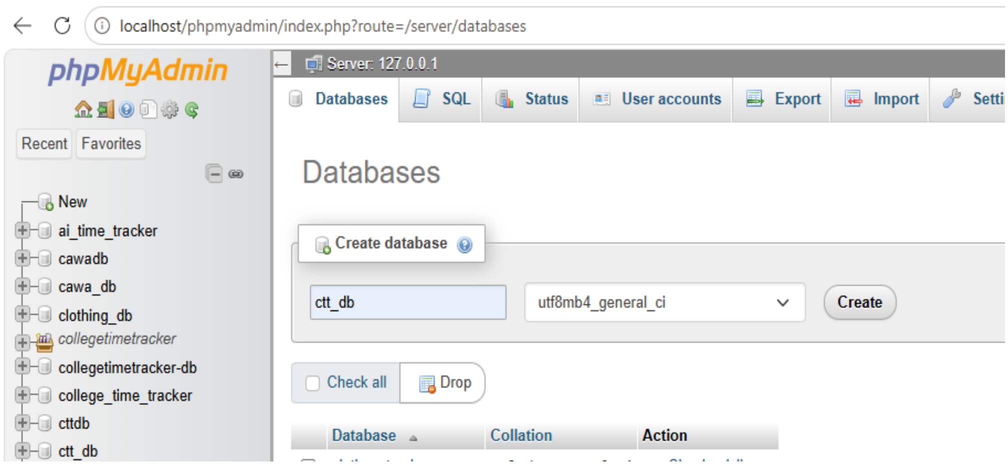
Step 1: Install XAMPP and start the Apache and MySQL services.



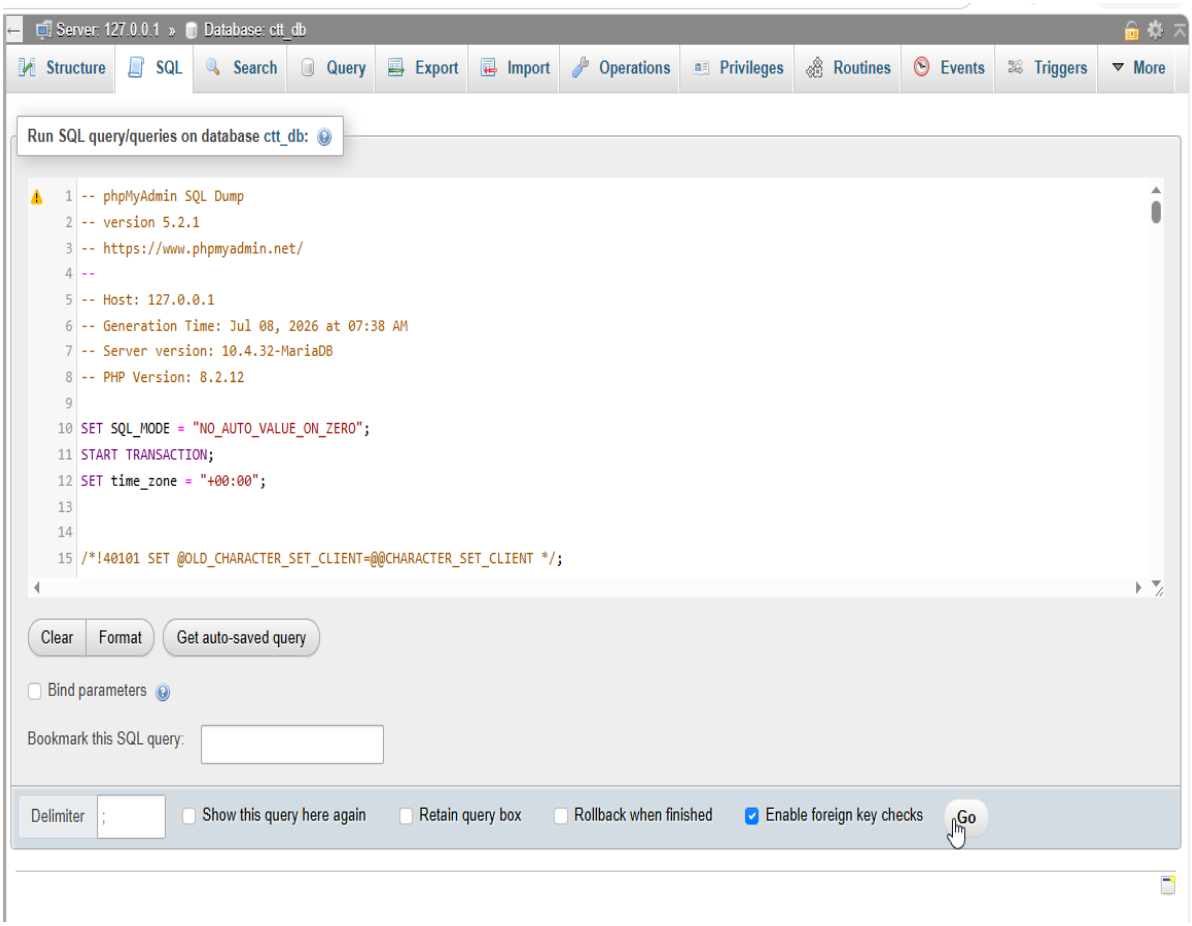
Step 2: Copy the project folder into C:\xampp\htdocs\ctt.



Step 3: Open phpMyAdmin and create a new database named ctt_db.



Step 4: Import the provided ctt_db.sql file into the newly created database.

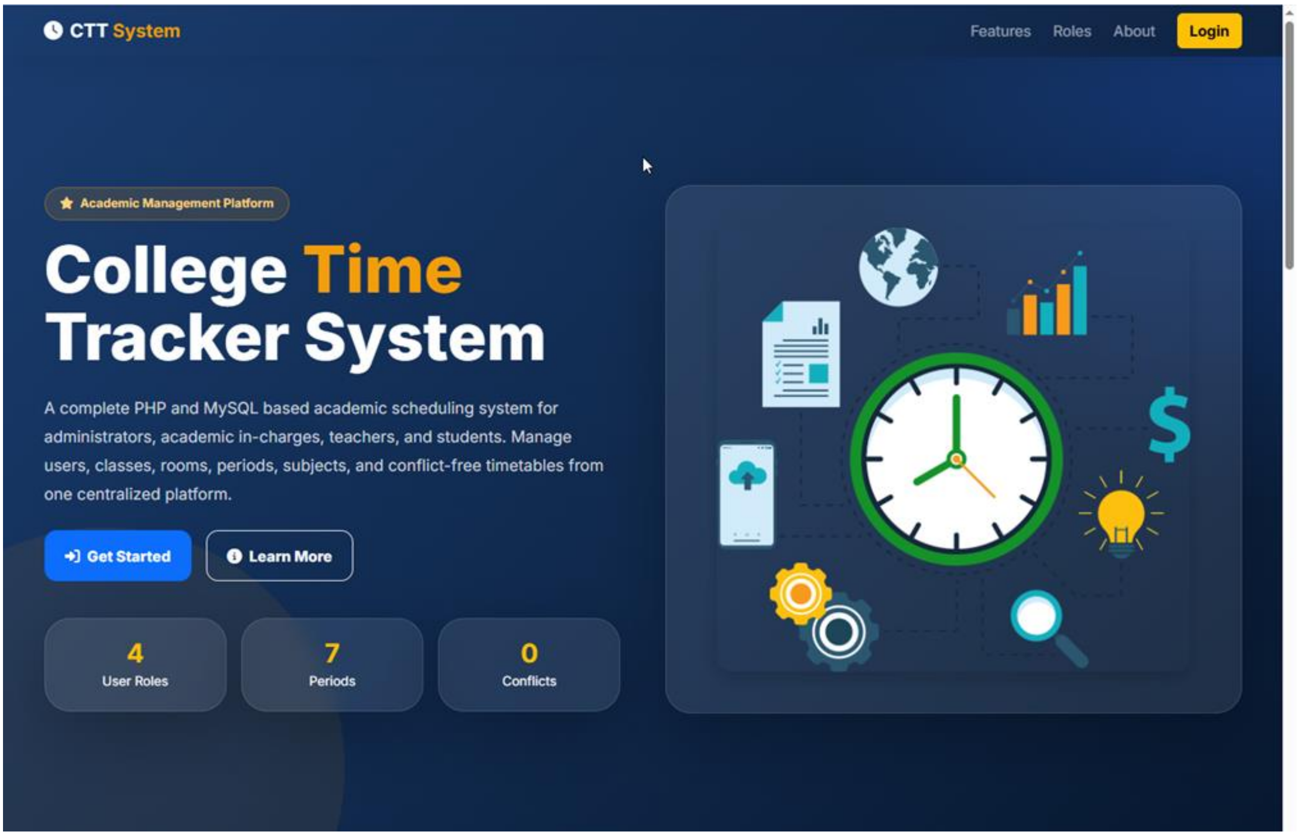


Step 5: Update the database connection credentials in includes/db.php.

ctt_php_project > includes > config.php

```php
<?php
// College Time Tracker - Main Configuration

define('APP_NAME', 'College Time Tracker');
define('APP_SHORT_NAME', 'CTT System');
define('BASE_PATH', dirname(__DIR__));

// XAMPP default database settings. Update password if your MySQL has one.
define('DB_HOST', 'localhost');
define('DB_USER', 'root');
define('DB_PASS', '');
define('DB_NAME', 'ctt_db');

define('APP_TIMEZONE', 'Asia/Karachi');
date_default_timezone_set(APP_TIMEZONE);

// In development keep errors visible. For final deployment, set to 0.
ini_set('display_errors', 1);
ini_set('display_startup_errors', 1);
error_reporting(E_ALL);

```

Step 6: Open a browser and navigate to http://localhost/ctt/index.php.



6.4 Deployment Details

- Hosting Environment: Local server (XAMPP) for development and testing; can be deployed to any shared hosting supporting PHP and MySQL for production use.

- Server Configuration: PHP display_errors disabled in production; database credentials stored in a protected configuration file.

6.5 User Manual

6.5.1 Introduction to the User Manual

This User Manual explains how to use the College Time Tracker (CTT) system in a simple, step-by-step manner. CTT is a web-based academic management platform used to:

- Create and manage class timetables
- Manage teachers, students, subjects, rooms, and periods
- Prevent scheduling conflicts automatically
- View personal schedules
- Generate workload reports
- Export and print timetables

Who Should Read This Manual?

Role	Why Read This Manual?
Administrator	Full system control, manage In-Charge accounts, backups, reports
Academic In-Charge	Build timetables, manage teachers/students, assign classes
Teacher	View personal teaching schedule
Student	View personal class timetable

How to Use This Manual

- Follow steps in order (Step 1 → Step 2 → Step 3).
- Words in bold are button names or menu labels.
- Words in code style are field names or URLs.
- Screenshots placeholders are shown as: [Figure X.X: Screen Name] (Insert actual screenshot here when printing the report.)

6.5.2 System Access Requirements

1 System URL

- Local (XAMPP): http://localhost/ctt/
- Live Server: https://ctt-webapp.hstn.me/

2 Default Demo Accounts (for testing)

Role	Username	Password
Administrator	admin	123456
Academic In-Charge	incharge	123456
Teacher	sajjad (or other teacher username)	123456
Student	ayesha (or other student username)	123456

Security Note: Change default passwords after first login in a real deployment.

3. Getting Started (Common for All Users)

3.1 Opening the Application

Step 1: Open your web browser.

Step 2: Type the CTT system URL in the address bar and press Enter.

Step 3: The Home Page (Landing Page) will appear.

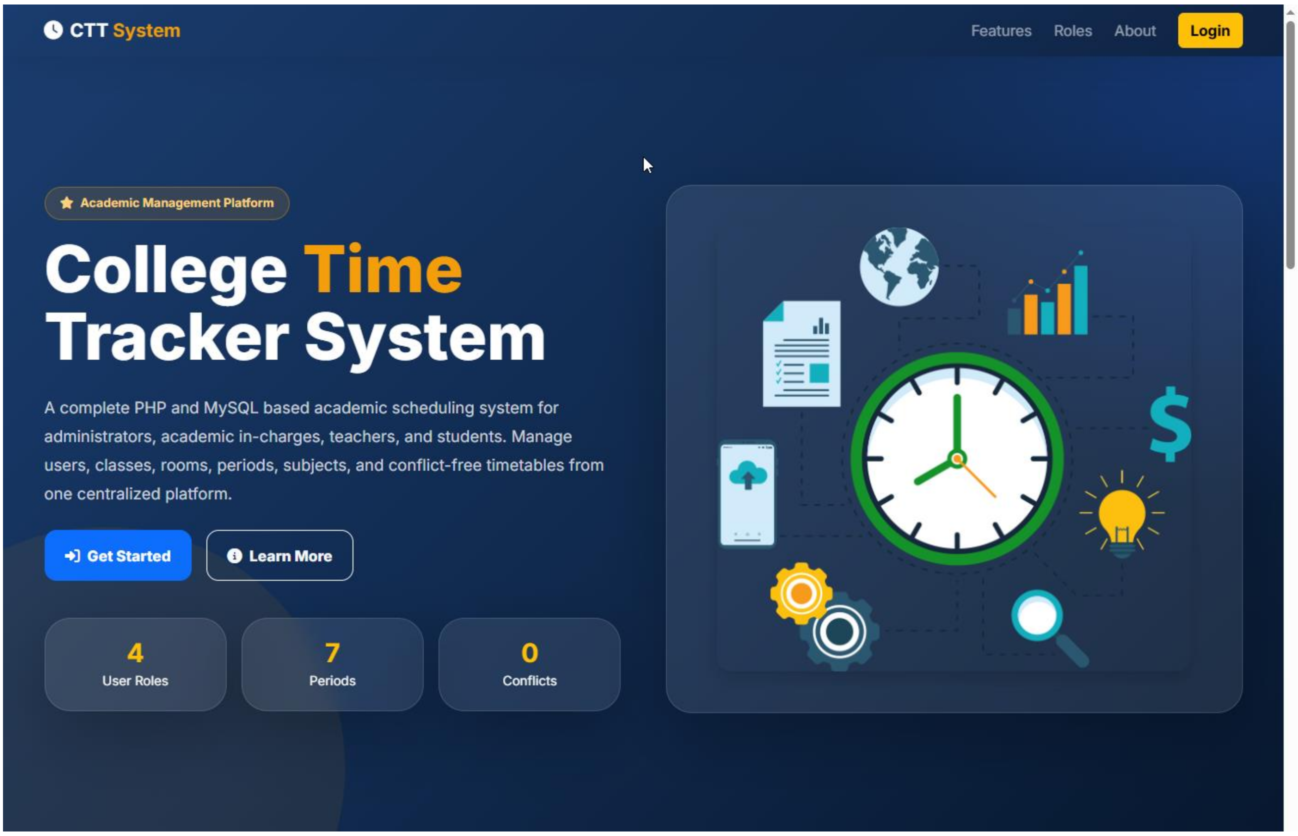


Figure 6.1: Home / Landing Page — Shows CTT logo, features section, user roles section, and Login button.

What You See on the Home Page

- System title: College Time Tracker (CTT)
- Short description of the platform
- Feature cards (Timetable Management, Role-Based Access, Reports, Conflict Prevention)
- Role cards (Administrator, Academic In-Charge, Teacher, Student)
- Top navigation: Home, Features, Roles, Login

3.2 Login Process (All Roles)

Step 1: Click the Login button on the top-right of the home page.

Step 2: The Login page opens.

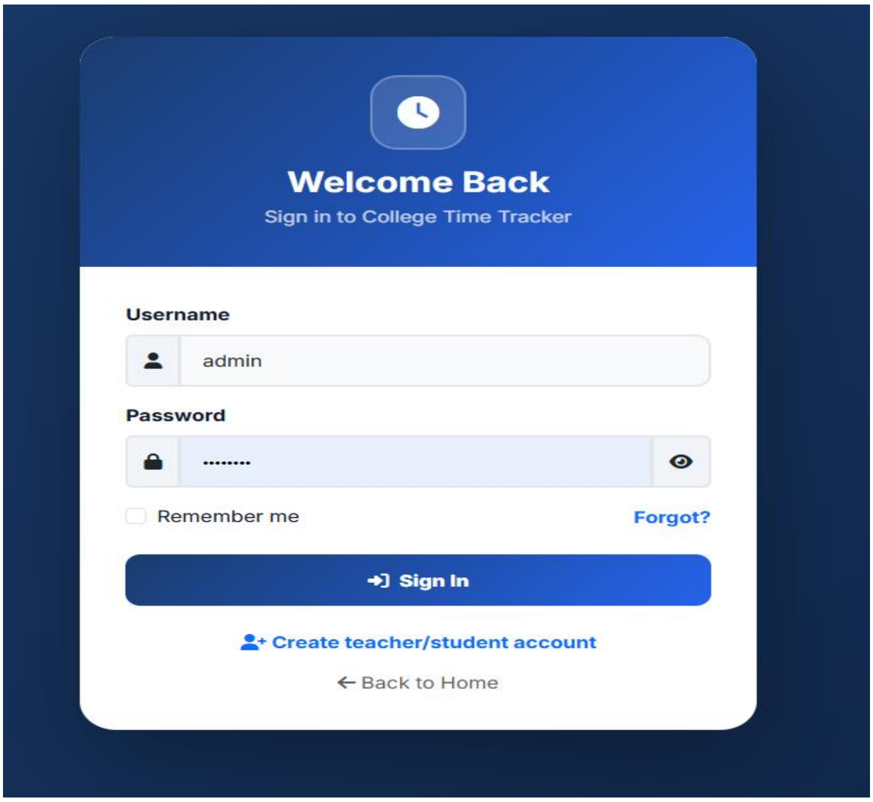


Figure 6.2: Login Page — Centered card with Username and Password fields, Remember Me checkbox, Forgot Password link, and Sign In button.

Step 3: Enter your Username.

Step 4: Enter your Password.

Step 5: (Optional): Tick Remember Me if you want the browser to keep you logged in longer.

Step 6: Click Sign In.

What Happens After Login?

The system checks your role and redirects you automatically:

Your Role	Redirect Destination
Administrator	Admin Dashboard
Academic In-Charge	In-Charge Dashboard
Teacher	Teacher Dashboard
Student	Student Dashboard

Login Errors and Solutions

Error Message	Cause	Solution
Please fill in all fields	Empty username or password	Fill both fields
Invalid username or password	Wrong credentials	Recheck username/password
Account blocked	Admin blocked your account	Contact Administrator

3.3 Logout Process (All Roles)

Step 1: On any dashboard, look at the left Sidebar.

Step 2: Click Logout at the bottom of the sidebar.

Step 3: You will be redirected to the Login page.

Step 4: Your session is destroyed (secure logout).

Always logout when using a shared computer.

3.4 Forgot Password

Step 1: On the Login page, click Forgot Password?.

Step 2: Enter your registered Email Address.

Step 3: Select your Role (Admin / In-Charge / Teacher / Student).

Step 4: Click Send Reset Instructions.

Step 5: System confirms that reset instructions were sent (demo mode shows success message).

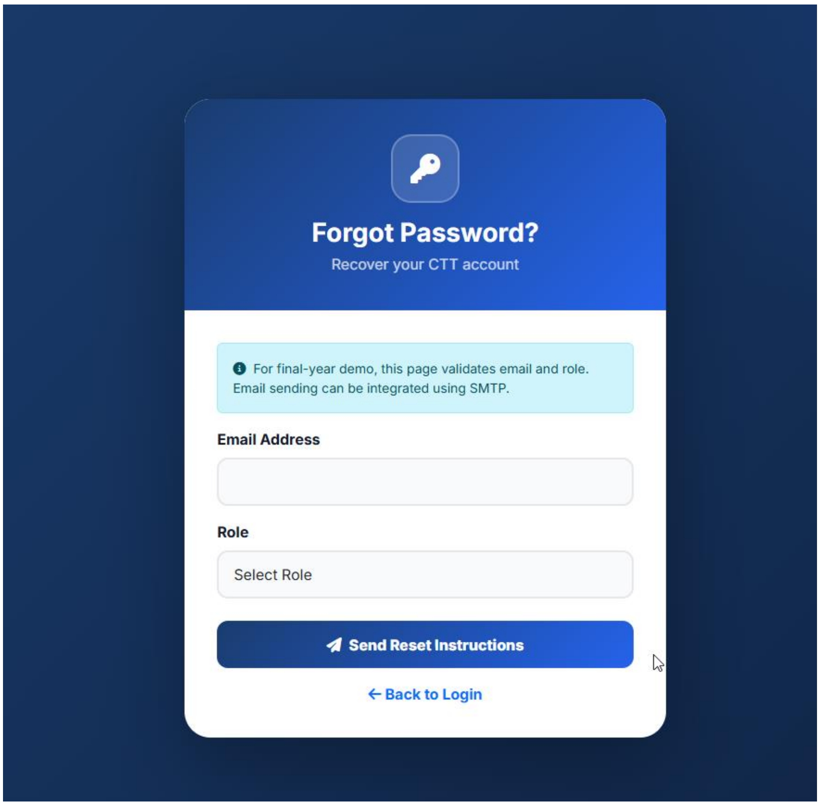


Figure 6.3: Forgot Password Page

3.5 Sign Up / Registration (Teachers & Students Only)

Administrator and Academic In-Charge accounts are created by the institution. Teachers and Students may self-register if the college allows it.

Step 1: On Login page, click Create New Account.

Step 2: Select role: Teacher or Student.

Step 3: Fill the form:

Field	Required?	Example
Full Name	Yes	Muhammad Ali
Email Address	Yes	ali@college.edu.pk
Department	No	Computer Science
Designation (Teacher only)	No	Lecturer
Roll Number (Student only)	Yes	BSCS-F22-001
Semester (Student only)	No	4th Semester
Username	Yes	ali_teacher
Password	Yes	Min 6 characters

Confirm Password	Yes	Same as password
Terms & Conditions	Yes	Must tick checkbox

Step 4: Click Create My Account.

Step 5: Success screen appears → click Proceed to Login.

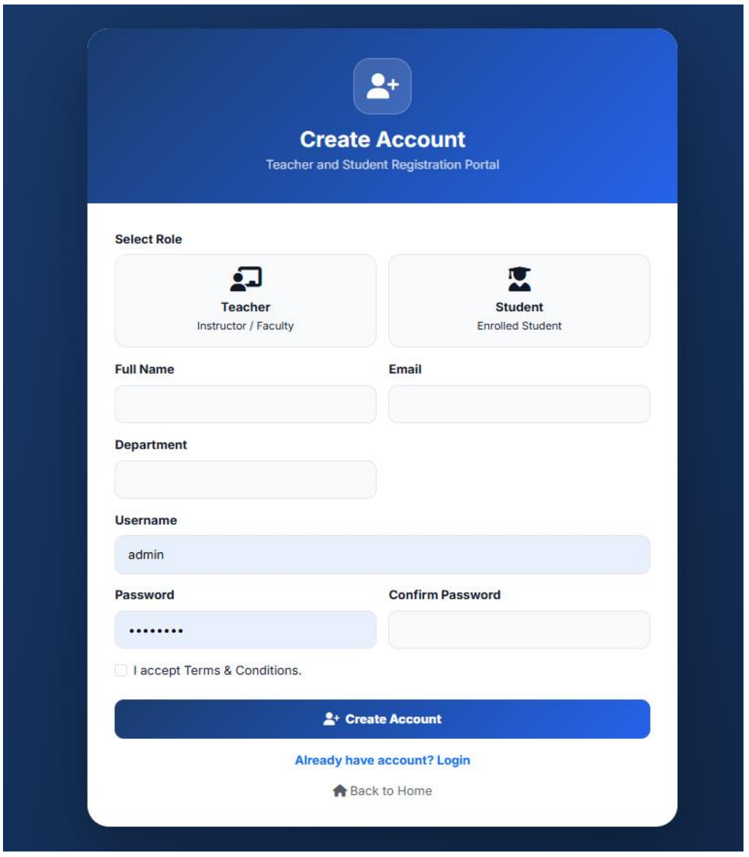


Figure 6.4: Sign Up Page

4. Administrator User Manual

Administrator has full system control.

4.1 Admin Dashboard Overview

After login, you land on the Admin Dashboard.

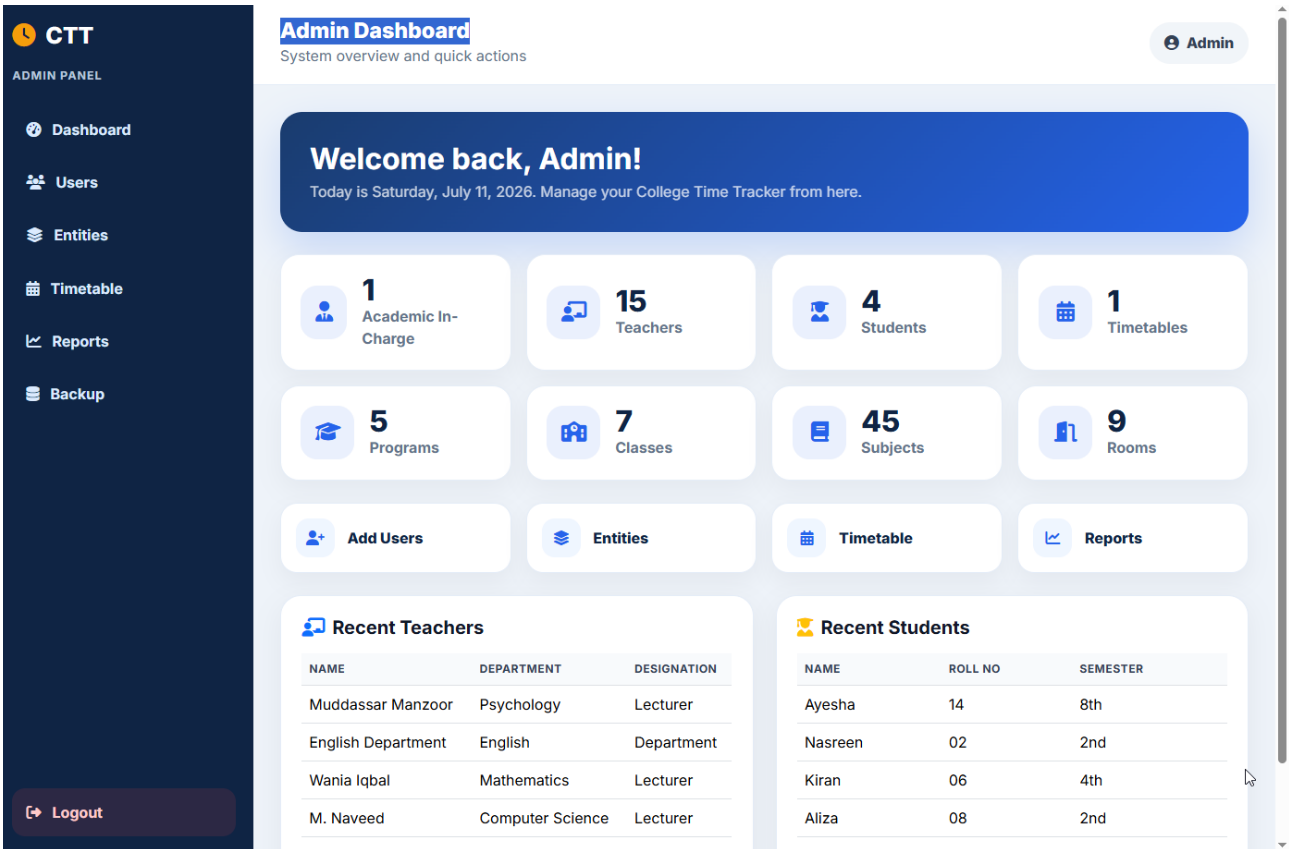


Figure 6.5: Admin Dashboard — Shows welcome banner, statistics cards, quick action buttons, recent teachers list, recent students list.

Dashboard Sections Explained

1. Welcome Banner

- Shows your name and today's date
- Example: Welcome back, Admin! Today is Monday, March 16, 2026

2. Statistics Cards (System Overview)

Card	Meaning
Academic In-Charge	Total In-Charge accounts
Teachers	Total teacher accounts
Students	Total student accounts
Timetables	Total timetable records
Programs	Total academic programs
Classes	Total classes
Subjects	Total subjects
Rooms	Total rooms

3. Quick Actions

Click any card to jump directly:

- Add In-Charge
- Add Teacher
- Add Student
- Add Program
- Add Class
- Add Subject
- Add Room
- Timetable Builder

4. Recent Activity Tables

- Recent Teachers (latest 5)
- Recent Students (latest 5)

4.2 Admin Sidebar Menu

Menu Item	Purpose
Dashboard	System overview

Academic In-Charge	Manage In-Charge accounts
Teachers	Manage teacher accounts
Students	Manage student accounts
Programs	Manage academic programs
Classes	Manage classes
Subjects	Manage subjects
Sections	Manage sections
Rooms	Manage rooms
Periods	Manage time periods
Timetable Builder	Create/edit timetable entries
View Timetable	Full grid view + export/print
Reports	Workload and analytics
Logout	Secure logout

4.3 Manage Academic In-Charge Accounts (FR02)

View All In-Charge Accounts

Step 1: Sidebar → Academic In-Charge

Step 2: Table shows: Name, Username, Email, Department, Actions

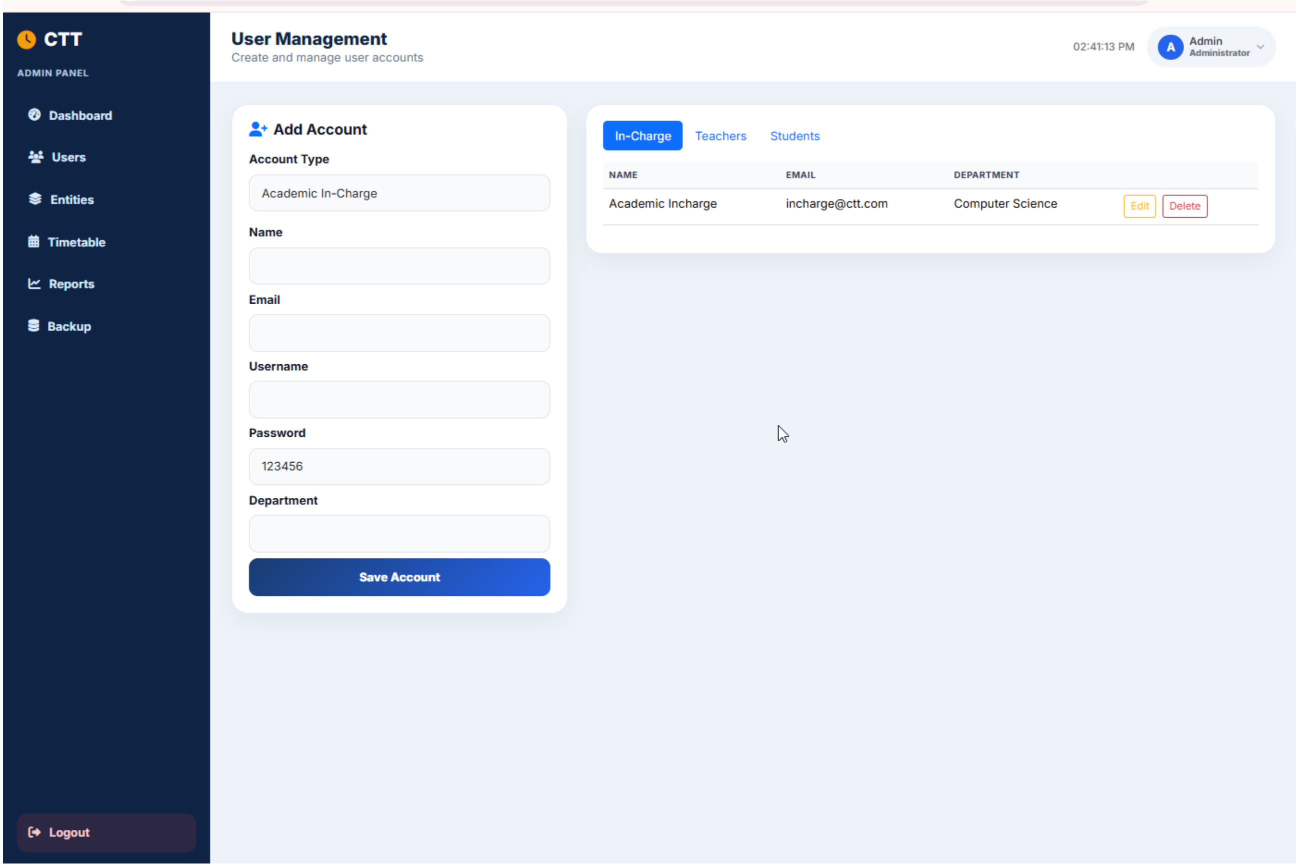


Figure 6.6: Manage In-Charge List

Add New In-Charge

Step 1: Click Add New button.

Step 2: Fill form:

Field	Example
Full Name	Ayesha Khan
Email	ayesha@ctt.edu.pk
Username	ayesha_incharge
Department	Computer Science
Password	123456 (or leave blank for default)

Step 3: Click Save.

Step 4: New In-Charge appears in the list.

Edit In-Charge

Step 1: Click the Edit (pen) icon next to a record.

Step 2: Update fields.

Step 3: Click Save. (Leave password blank if you do not want to change it.)

Delete In-Charge

Step 1: Click the Delete (trash) icon.

Step 2: Confirm the popup: Delete this record?

Step 3: Record is removed.

4.4 Manage Teachers (FR08 / Admin Level)

Path: Sidebar → Teachers

Same process as In-Charge:

- Add New → Name, Email, Username, Department, Designation, Password
- Edit → update details
- Delete → confirm and remove

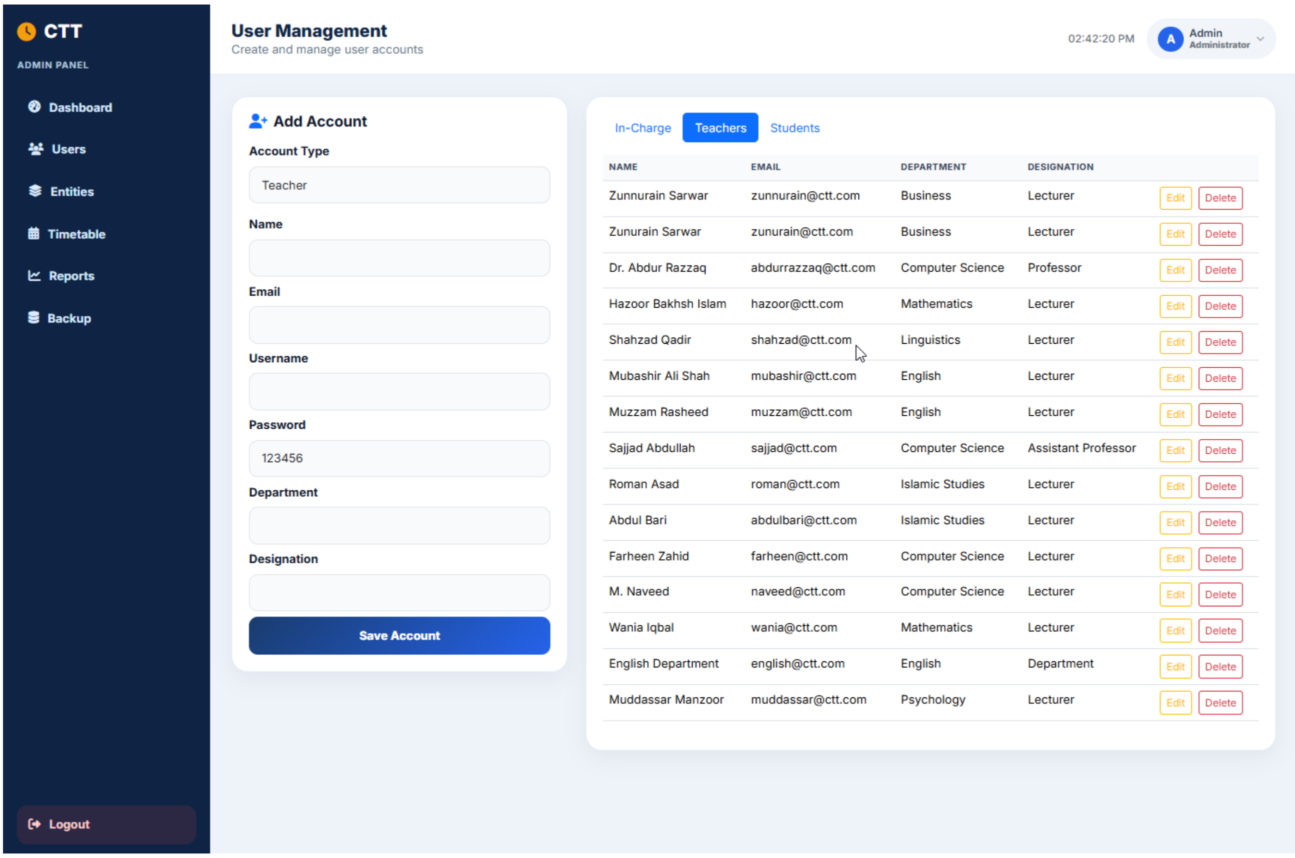


Figure 6.7: Manage Teachers Screen

4.3 Manage Students

Path: Sidebar → Students

Field	Notes
Full Name	Required
Roll Number	Required + Unique
Email	Required + Unique
Username	Required + Unique
Department	Optional
Semester	Select from dropdown (1st–8th)
Password	Default 123456 if blank

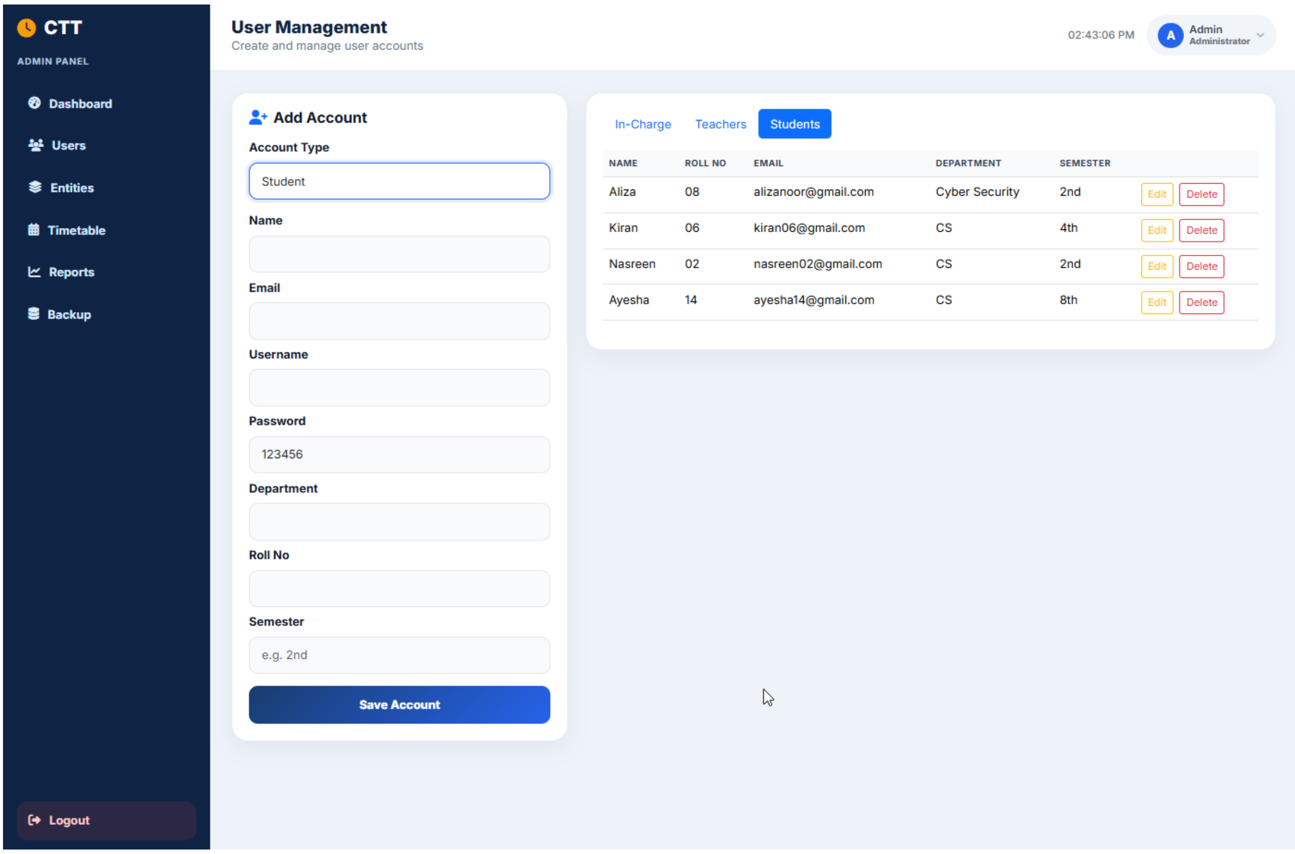


Figure 6.8: Manage Students Screen

4.6 Manage Academic Entities

1. Programs

Path: Sidebar → Programs

Field	Example
Program Name	BSCS
Academic System	Semester / Annual / Quarter

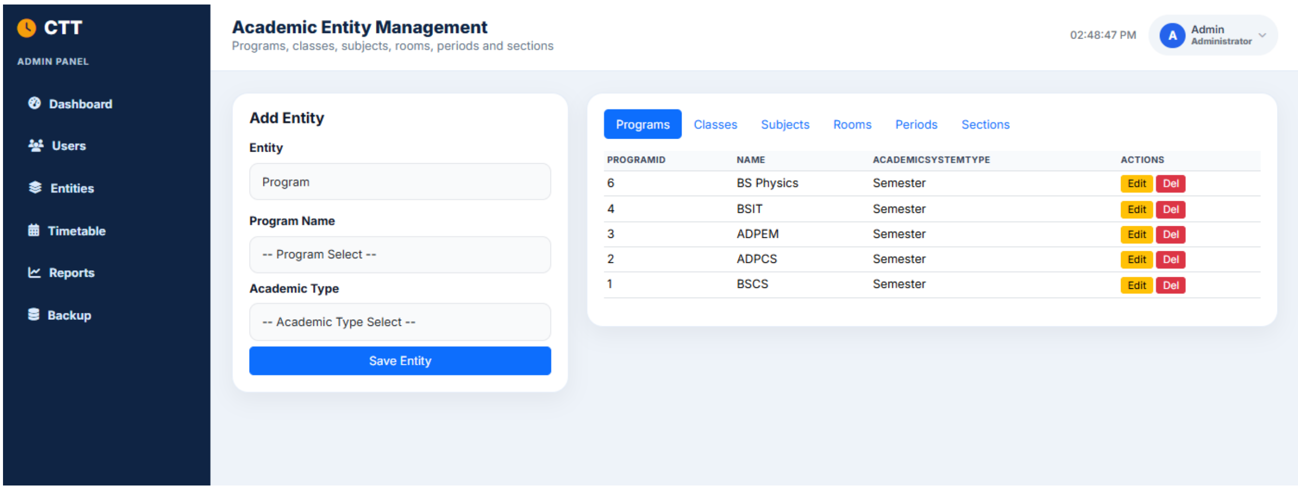


Figure 6.9: Manage Programs Screen

2. Classes

Path: Sidebar → Classes

Field	Example
Class Name	BSCS 2nd Semester
Session	2025-27
Current Semester	2nd
Timetable Shift	Morning Shift Timetable
Program	BSCS
Row Color	Green (for grid display)
Important Dates	Mid Term Exam 13.04.26 ...
Display Order	1
Active	Tick checkbox

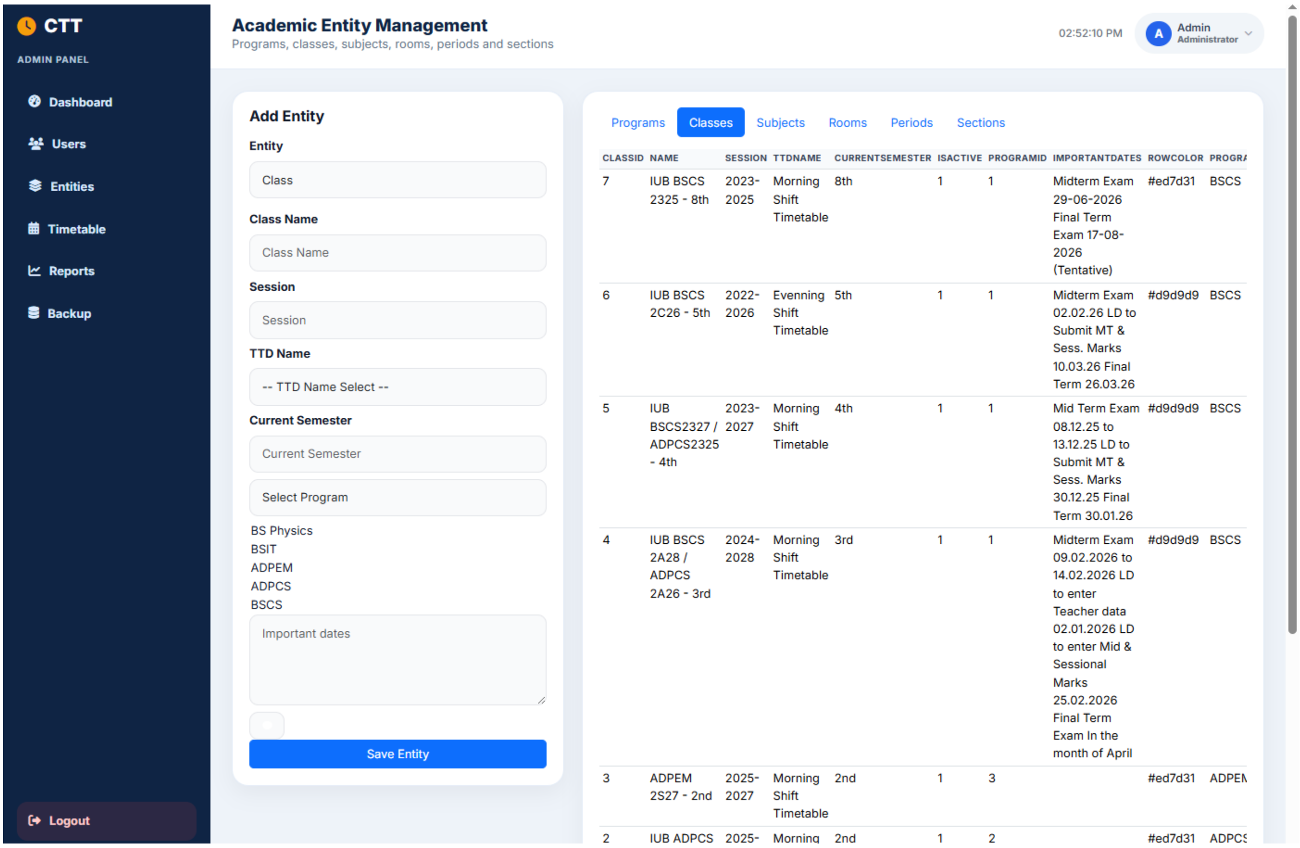


Figure 6.10: Manage Classes Screen

3. Subjects

Path: Sidebar → Subjects

Field	Example
Subject Name	Data Structures
Class	BSCS 2nd Semester
Semester	2nd Semester
Display Order	1

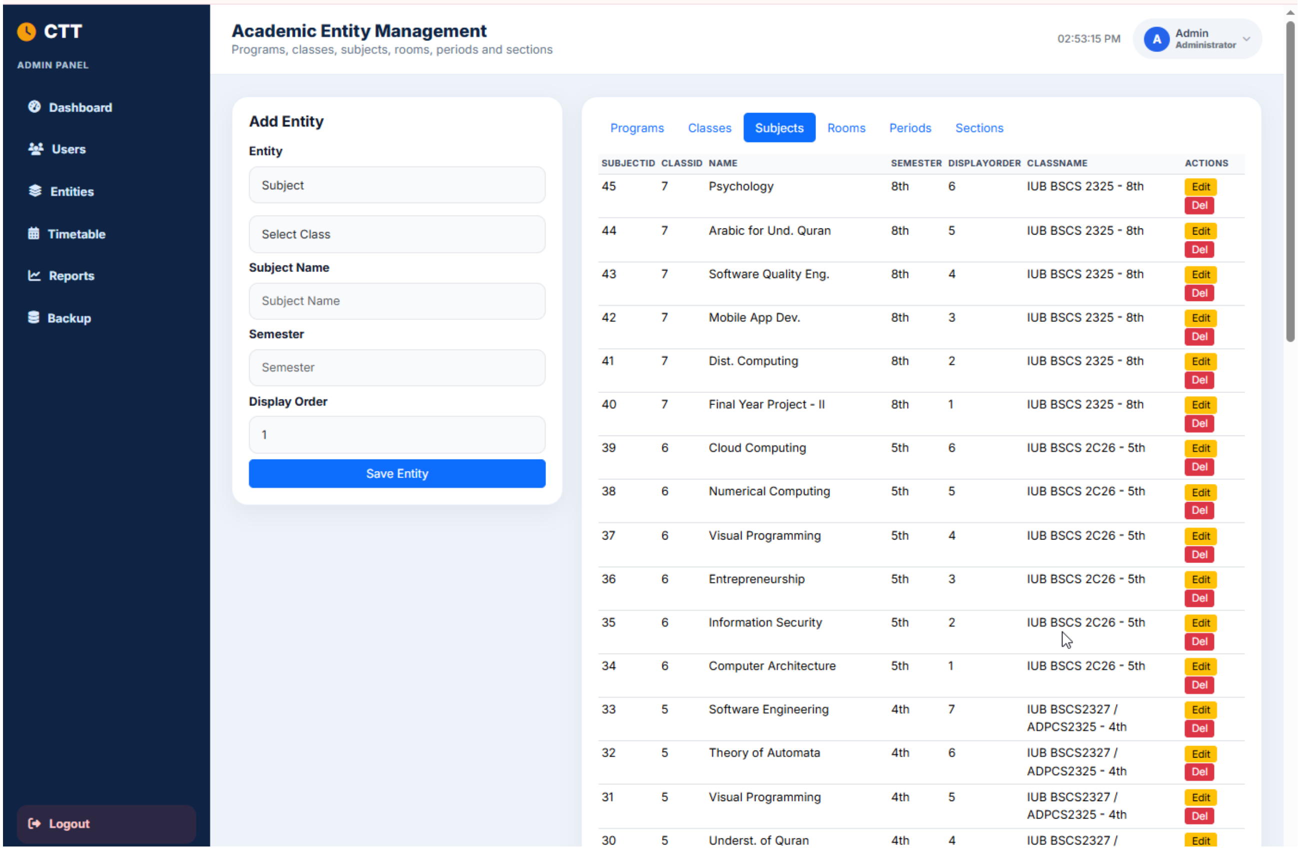


Figure 6.11: Manage Subjects Screen

4. Sections

Path: Sidebar → Sections

Field	**Example**
Section Name	A / Regular / Morning
Semester	2nd
Subject	Data Structures
Teacher	Sajjad Abdullah
Active	Yes

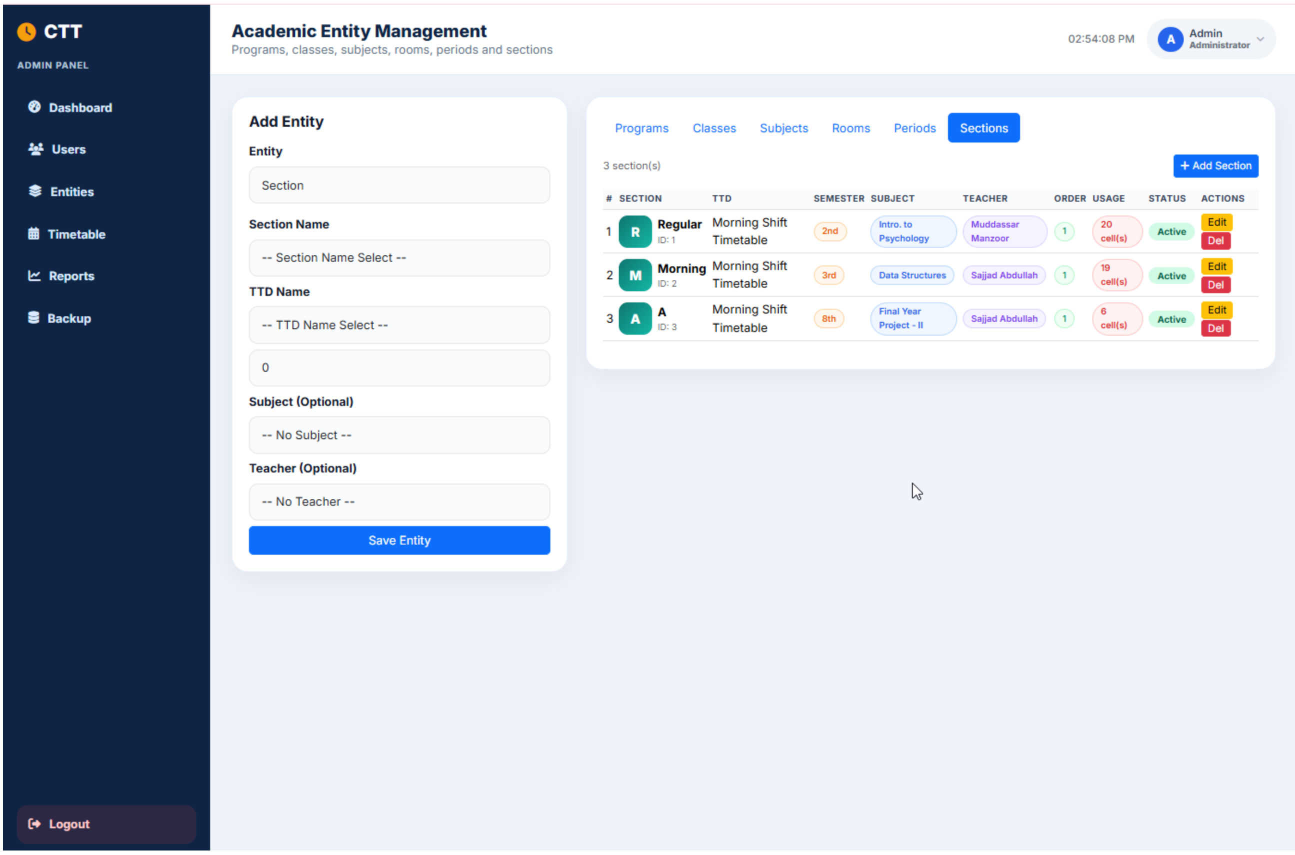


Figure 6.12: Manage Sections Screen

5. Rooms

Path: Sidebar → Rooms

Field	**Example**
Room Title	Room No. 15 / Comp. Lab 1
Timetable Shift	Morning Shift Timetable

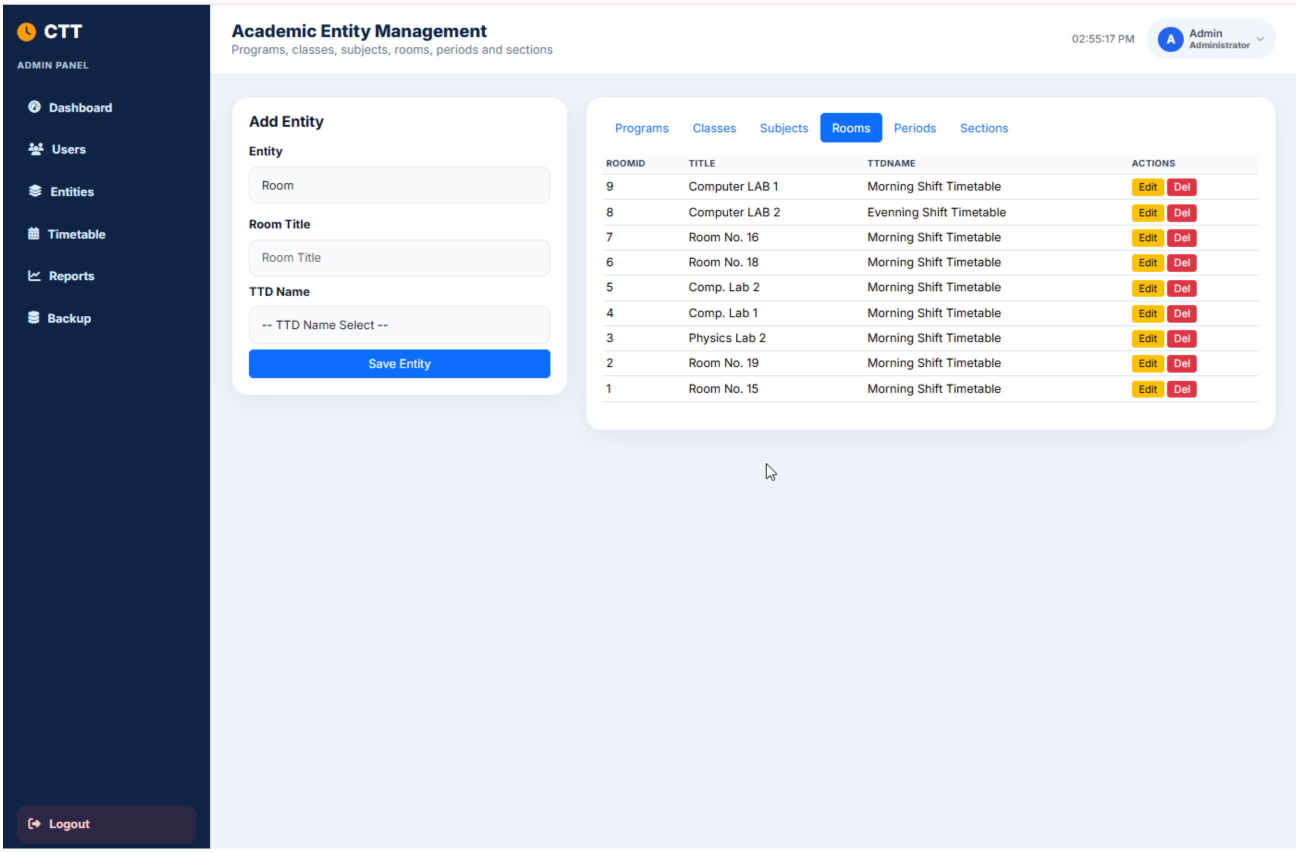


Figure 6.13: Manage Rooms Screen

6. Periods

Path: Sidebar → Periods

Field	Example
Title	Period 1
Number	1
Start Time	08:30
End Time	09:20
Friday Start	08:30
Friday End	09:15
Display Order	1

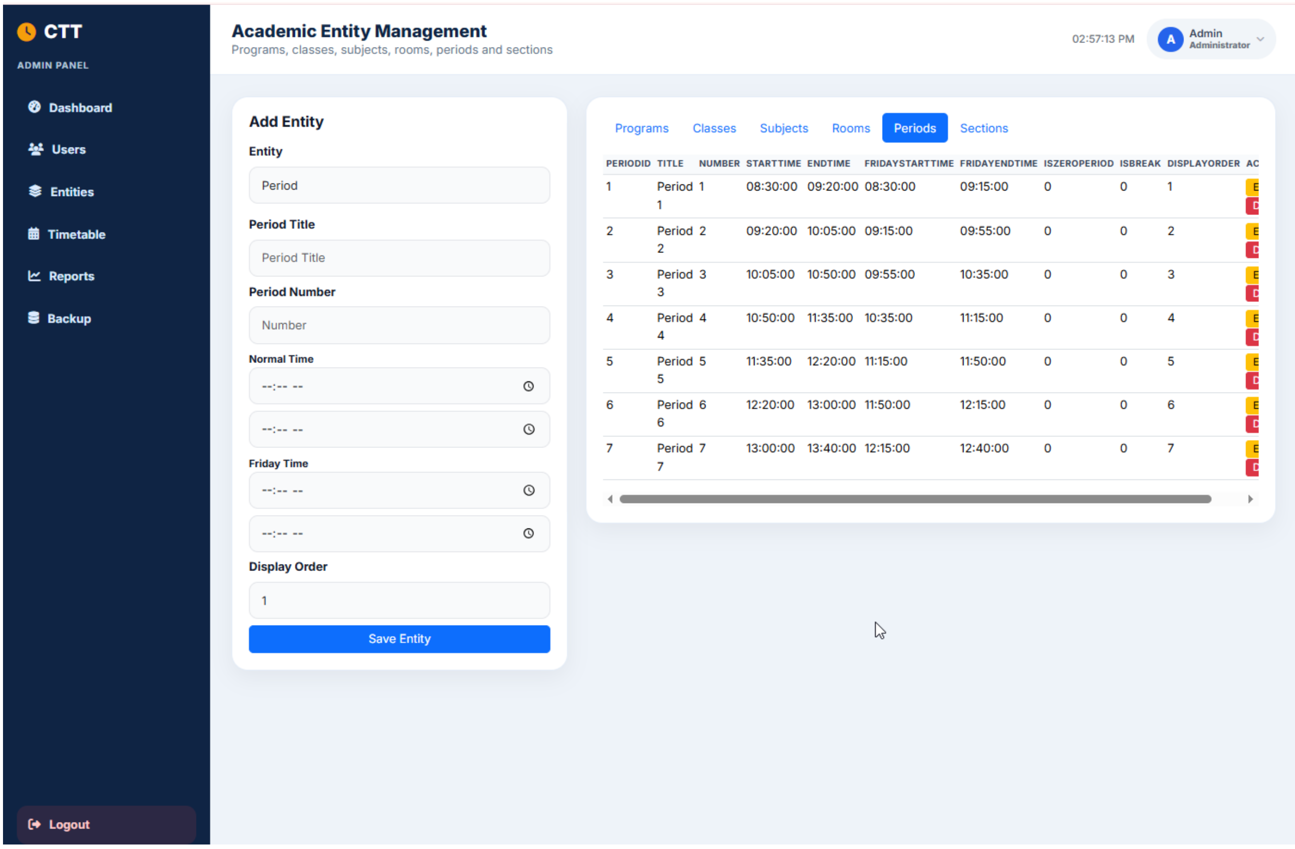


Figure 6.14: Manage Periods Screen

4.7 Timetable Builder (Most Important Admin Feature)

Path: Sidebar → Timetable Builder

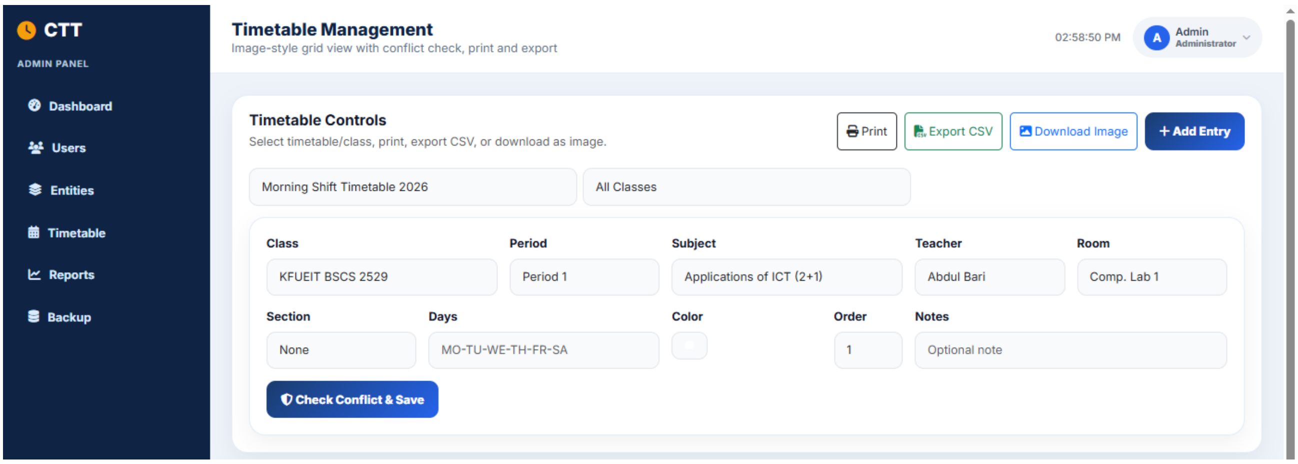


Figure 6.15: Timetable Builder Page — Top form to add entry + list of all entries + live grid preview at bottom.

How to Add a Timetable Entry

Step 1: Open Timetable Builder.

Step 2: In the "Add Timetable Entry" form, select:

Field	Example
Class *	BSCS 2nd Semester
Period *	P1 (08:30-09:20)
Subject	Intro. to Psychology
Teacher	Mudassar Manzoor
Room	Room No. 15
Section	Regular
Days	MO-TU

Step 3: Click Add Entry.

What Happens Internally?

- System checks if the teacher is free in that period on those days.
- If no conflict → entry is saved and grid updates.
- If conflict → red error appears: "Conflict! This teacher is already assigned to another class in this period on overlapping days."

View All Entries Table

Below the form you will see a table of all saved entries with:

- Class, Period, Subject, Teacher, Room, Days
- Conflict badge (if any)
- Delete button for each entry

Live Preview Grid

At the bottom, the full timetable grid updates automatically after every add/delete.

Delete an Entry

Step 1: In the entries table, click the trash icon.

Step 2: Entry is removed and grid refreshes.

4.8 View Full Timetable (Export & Print)

Path: Sidebar → View Timetable

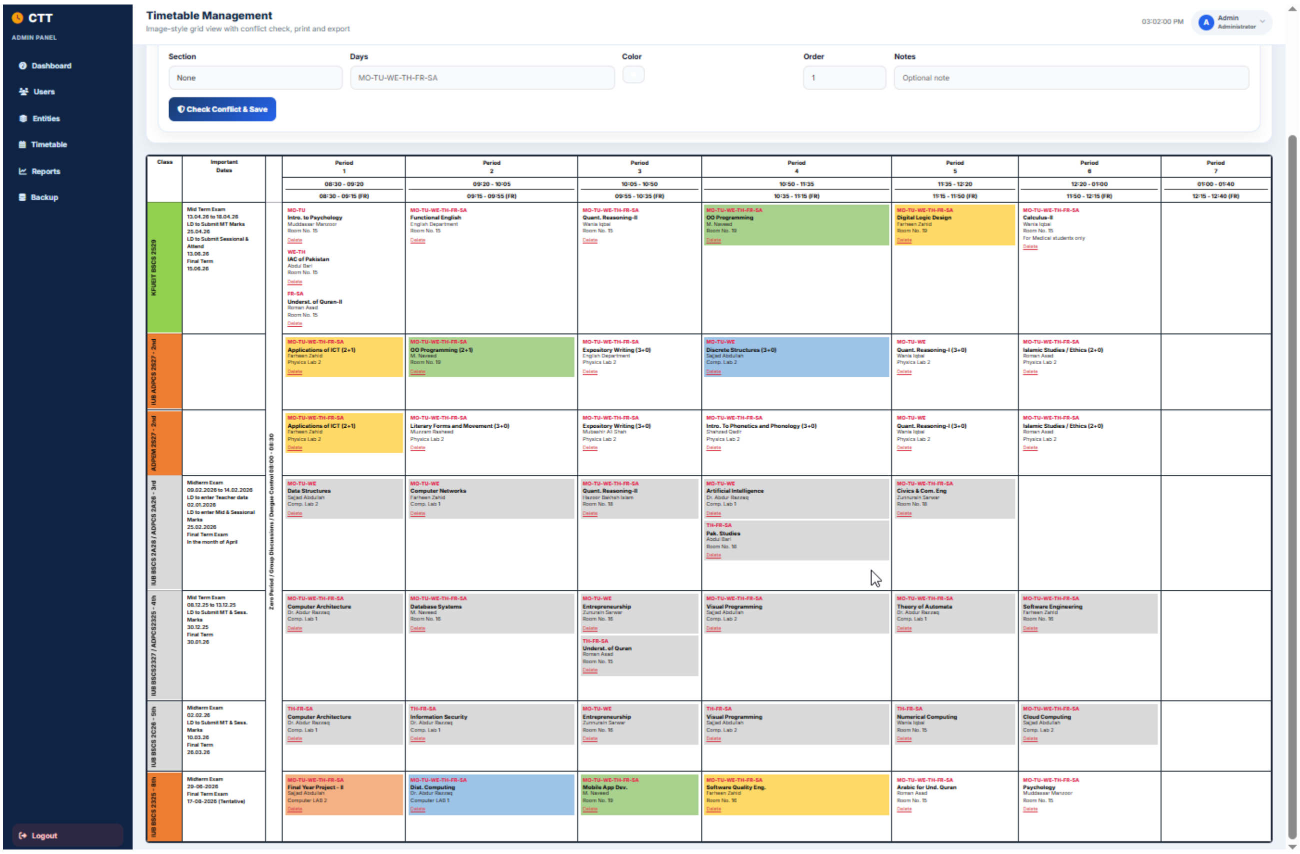


Figure 6.16: Full Timetable Grid View — Class rows × Period columns, Important Dates column, colored class labels, subject/teacher/room in cells.

Toolbar Buttons

Button	Action
Print	Opens browser print dialog (landscape recommended)
Export CSV	Downloads timetable.csv file

Conflict Banner

- Green badge: No conflicts
- Red badge: X conflict(s) detected + list of conflicting teachers

How to Print Professionally

Step 1: Click Print.

Step 2: In print settings:

- Layout: Landscape
- Margins: Minimum
- Background graphics: On (optional for colors)

Step 3: Click Print or Save as PDF.

4.9 Reports Module

Path: Sidebar → Reports

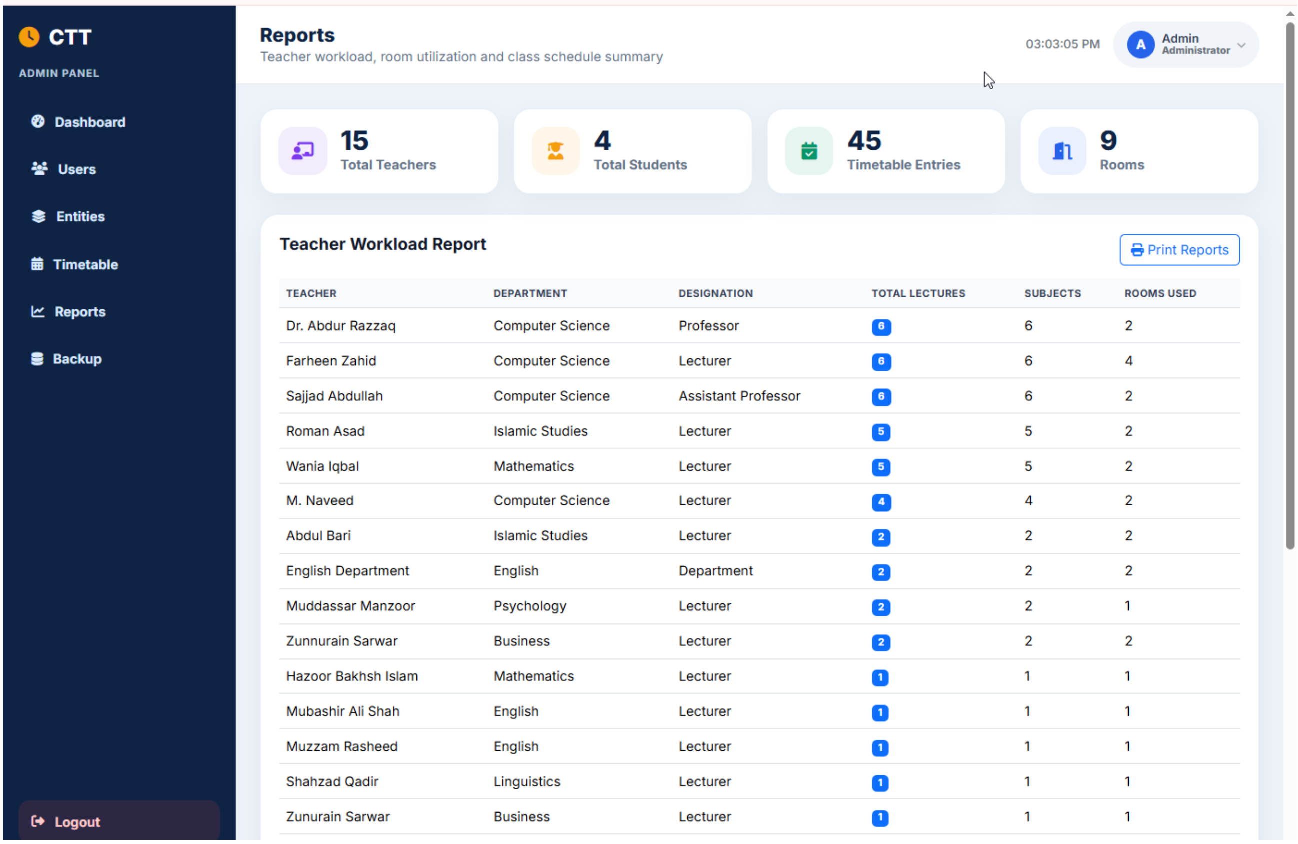


Figure 6.17: Reports Page

Available Reports

1. **Teacher Workload Report**

Columns:

- Teacher Name
- Department
- Designation
- Number of Classes Taught
- Weekly Periods Assigned

2. **Class Load Report**

Columns:

- Class Name
- Distinct Subjects
- Scheduled Periods

3. **Conflict Summary**

- Total conflicts detected
- Detailed conflict messages

Print Report: Click Print Report button → browser print dialog.

4.10 Database Backup / Restore (Conceptual)

Path: (If implemented) Sidebar → Database Backup

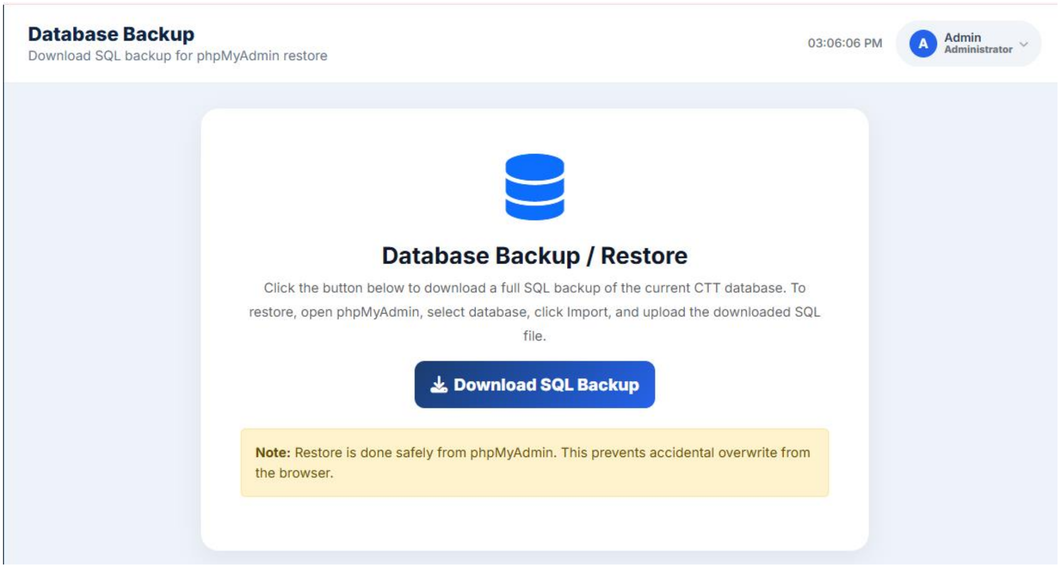


Figure 6.18: Database Backup / Restore Page Screen

Backup Steps:

- Click Download Backup.
- SQL file downloads to your computer.
- Store it safely.

Restore Steps:

- Open phpMyAdmin.
- Select ctt_db.
- Import the SQL backup file.

5. Academic In-Charge User Manual

Academic In-Charge manages day-to-day academic operations.

5.1 In-Charge Dashboard

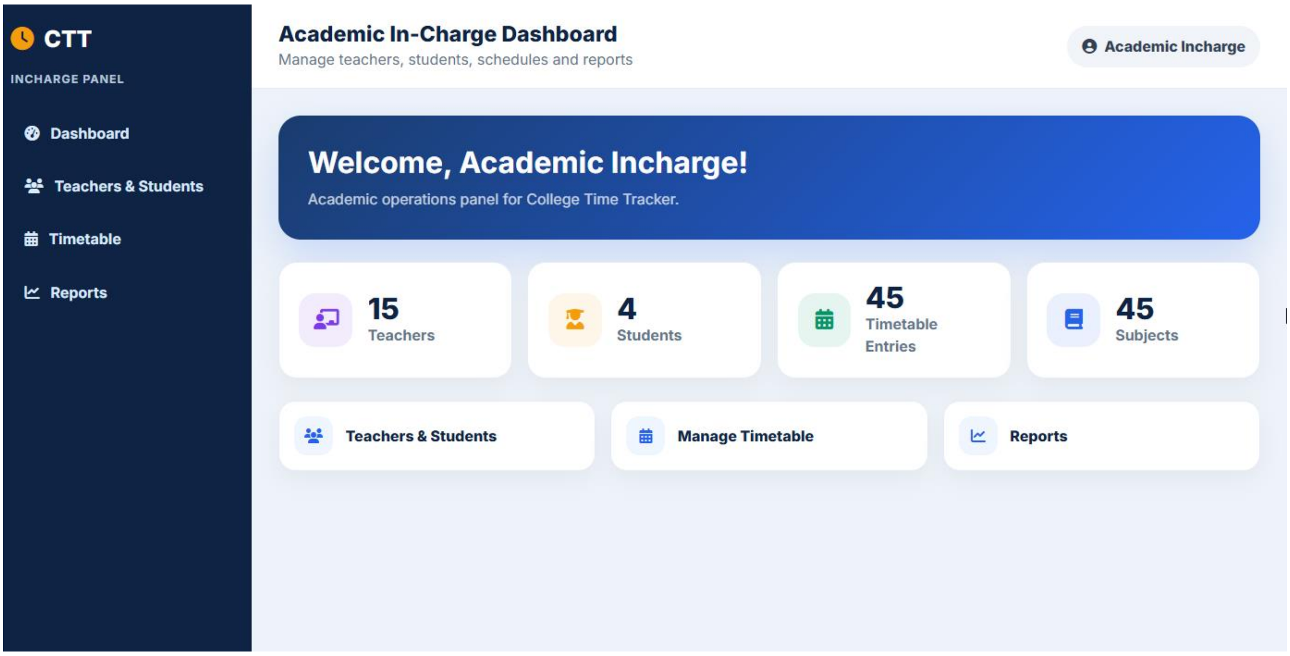


Figure 6.19: In-Charge Dashboard — Welcome banner (green theme), stats for Teachers/Students/Subjects/Timetables, quick actions, recent students table.

Sidebar Menu (In-Charge)

Menu Item	Purpose
Dashboard	Overview
Teachers	Manage teachers
Students	Manage students
Subjects	Manage subjects
Sections	Manage sections
Timetable Builder	Build schedule
View Timetable	Full grid + export/print
Reports	Workload reports
Logout	Exit

5.2 What In-Charge Can Do

Feature	Allowed?
Manage Teachers	Yes
Manage Students	Yes
Manage Subjects / Sections	Yes
Build Timetable	Yes
View / Export / Print Timetable	Yes
View Reports	Yes
Manage In-Charge accounts	No (Admin only)
Database Backup	No (Admin only)

5.3 Typical Daily Workflow for In-Charge

Step 1: Login → Dashboard

Step 2: Add new teachers/students if needed

Step 3: Ensure subjects and rooms are complete

Step 4: Open Timetable Builder

Step 5: Add entries carefully (watch for conflict messages)

Step 6: Open View Timetable to verify grid

Step 7: Export CSV / Print for notice board

Step 8: Check Reports for teacher workload balance

Step 9: Logout

6. Teacher User Manual

Teachers have a simple, read-focused portal.

6.1 Teacher Dashboard

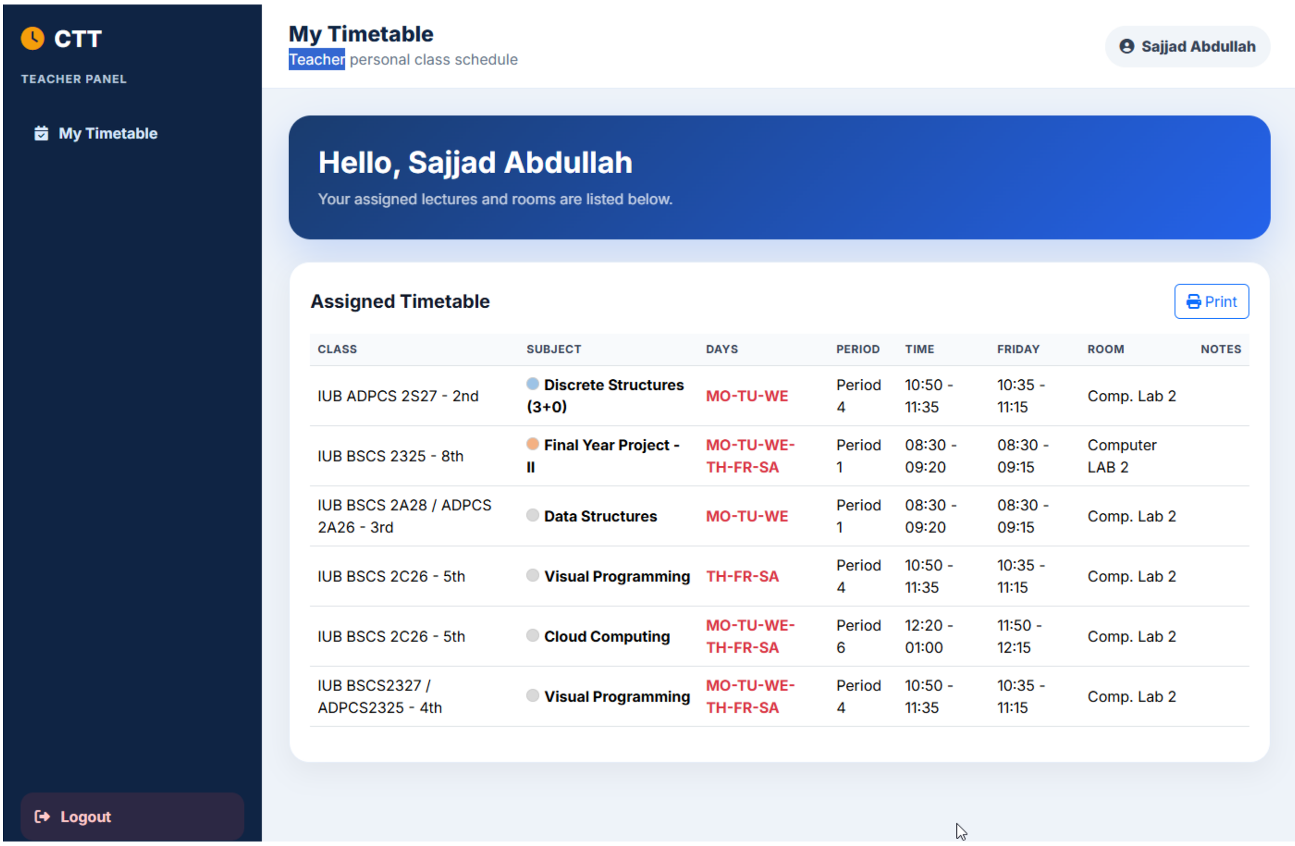


Figure 6.20: Teacher Dashboard — Purple welcome banner, stats (Weekly Classes, Classes Taught, Rooms), My Teaching Schedule table.

What Teachers See

Section	Description
Welcome Banner	Name + Designation + Department
Weekly Classes	Total assigned periods
Classes Taught	Number of distinct classes
Rooms	Number of rooms used
My Teaching Schedule	Detailed table of all assignments

My Teaching Schedule Table Columns

Column	Meaning
Class	Which class you teach
Subject	Subject name
Days	e.g., MO-TU-WE
Period	e.g., P2 (09:20-10:05)
Room	Room number

6.2 View Full Timetable

Path: Sidebar → Full Timetable

- Full college grid opens
- Your classes are highlighted (other entries appear dimmed)
- You can still Print / Export CSV

6.3 What Teachers Cannot Do

- Cannot add/edit/delete timetable entries
- Cannot manage users
- Cannot change other teachers' schedules

7. Student User Manual

Students have the simplest interface.

7.1 Student Dashboard

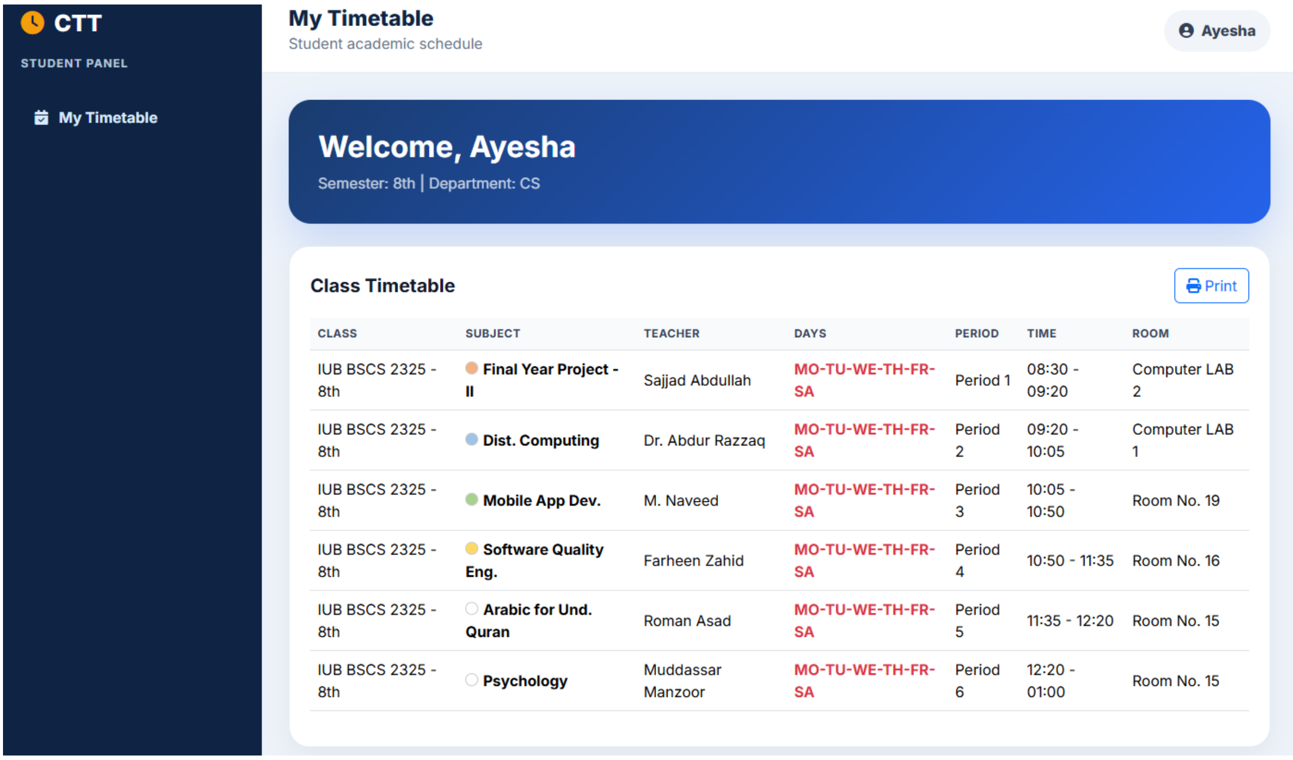


Figure 6.21: Student Dashboard — Orange/gold welcome banner, Roll No / Department / Semester cards, class timetable grid.

What Students See

Card	Content
Roll Number	e.g., 14
Department	e.g., CS
Semester	e.g., 8th Semester
Class Timetable	Grid filtered/highlighted for their semester

7.2 How Student Timetable Matching Works

Step 1: System reads student's Semester field.

Step 2: Matches it with class Current Semester.

Step 3: Highlights that class row in the grid.

Step 4: Other classes appear dimmed.

7.3 View Full Timetable

Path: Sidebar → Full Timetable. Same as teacher — full grid, print, export.

7.4 What Students Cannot Do

- Cannot edit anything
- Cannot see other students' personal data
- Cannot access admin/in-charge modules

8. Common Features (All Roles)

8.1 Responsive Design (Mobile / Tablet)

- On mobile, sidebar collapses into a hamburger menu (☰) top-left.
- Tap ☰ to open/close navigation.
- Timetable grid scrolls horizontally on small screens.

8.2 Print Feature

Works on: Full Timetable page and Reports page.

Print tips:

- Always choose Landscape orientation
- Enable background graphics if you want colors
- Use "Save as PDF" to create soft copies for WhatsApp/email

8.3 CSV Export

Step 1: Open Full Timetable.

Step 2: Click Export CSV.

Step 3: File timetable.csv downloads.

Step 4: Open in Excel / Google Sheets.

8.4 Session Security

- Session expires after inactivity (or browser close, depending on settings).
- Always logout on shared computers.

- Passwords are never shown in plain text.

9. Troubleshooting Guide

Problem	Possible Cause	Solution
Cannot login	Wrong username/password	Reset password / contact admin
Page not loading	Server down / wrong URL	Check XAMPP Apache/MySQL are running
Conflict error while saving	Teacher already booked	Choose different teacher, period, or days
Timetable grid empty	No entries added yet	Use Timetable Builder to add entries
Student sees wrong class	Semester not set correctly	Admin/In-Charge updates student's semester
Print cuts off columns	Portrait mode used	Switch to Landscape
CSV not downloading	Browser blocked download	Allow downloads for the site
Sidebar not visible on phone	Menu collapsed	Tap hamburger (☰) icon
"Email already exists"	Duplicate registration	Use another email or login instead
Session expired suddenly	Long inactivity	Login again

10. Frequently Asked Questions (FAQs)

Q1: Who can create Administrator accounts?

A: Only an existing Administrator (or database seed script during installation).

Q2: Can a Teacher edit the timetable?

A: No. Only Administrator and Academic In-Charge can edit.

Q3: What does "MO-TU-WE-TH-FR-SA" mean?

A: Days of the week: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday.

Q4: Can one cell have multiple subjects?

A: Yes. Example: Period 1 of BSCS 2nd may have MO-TU Psychology and WE-TH IAC of Pakistan.

Q5: How does conflict detection work?

A: System checks if the same teacher is assigned to a different class in the same period on any overlapping day. If yes, save is blocked.

Q6: Can I use CTT offline?

A: Only if hosted on a local college LAN. Fully offline (no server) is not supported.

Q7: How do I change my password?

A: Contact Admin/In-Charge to reset, or use profile update if available for your role.

Q8: Is my data safe?

A: Passwords are hashed. Role-based access prevents unauthorized module entry. Use HTTPS on live servers for full safety.

Q9: Can multiple In-Charge users work at the same time?

A: Yes, but they should coordinate to avoid editing the same timetable simultaneously.

Q10: What file formats can I export?

A: CSV (Excel) and Print-to-PDF (browser print).

11. Glossary of Terms

Term	Meaning
CTT	College Time Tracker
RBAC	Role-Based Access Control
Period	A fixed time slot (e.g., 08:30–09:20)
Class	Academic group (e.g., BSCS 2nd Semester)
Section	Sub-group of a class (A, B, Regular)
Conflict	Overlapping assignment of teacher/room
Dashboard	Home screen after login for each role

Grid	Visual matrix of Class rows × Period columns
CSV	Comma-Separated Values (Excel-compatible file)
Session	Secure login state stored in browser cookie
CRUD	Create, Read, Update, Delete operations
In-Charge	Academic In-Charge (timetable coordinator)

12. Quick Reference Card (One-Page Summary)

Login

Home → Login → Username + Password → Sign In

Admin Core Tasks

Dashboard → Users/Entities → Timetable Builder → View Timetable → Reports

In-Charge Core Tasks

Dashboard → Teachers/Students → Timetable Builder → Export/Print

Teacher Core Tasks

Dashboard → My Teaching Schedule → Full Timetable (highlighted)

Student Core Tasks

Dashboard → My Class Timetable → Full Timetable

Emergency Contacts

- Technical Support: admin@ctt.edu.pk
- Academic Coordination: incharge@ctt.edu.pk

CHAPTER 7:

CONCLUSION AND

FUTURE ENHANCEMENT

7.1 Conclusion

The College Time Tracker system successfully achieves its primary objective of automating the academic timetable creation process while preventing scheduling conflicts through role-based access and an automated conflict-detection engine. All 17 functional requirements identified during the requirements analysis phase were implemented and verified through structured black-box testing, achieving a high pass rate. The system evaluation confirms that CTT significantly reduces the manual effort involved in timetable preparation and provides teachers and students with immediate, accurate access to their personal schedules.

7.2 Limitations

- The system does not currently support multi-campus deployment.
- Notifications (SMS/Email) for timetable changes are not implemented.
- The system assumes a single academic session is active at a time.

7.3 Future Enhancements

- Integration of an SMS/Email notification module for schedule changes.
- Development of a native mobile application for Android and iOS.
- Cloud-based deployment with automated database backups.
- AI-based automatic timetable generation using constraint-satisfaction algorithms.
- Integration of biometric attendance with the timetable module.

CHAPTER 8:

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